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How Experiments with Children Inform Economics[†]

JOHN A. LIST, RAGAN PETRIE, AND ANYA SAMEK*

In the past several decades, the experimental method has lent deep insights into economics. One area that has contributed is the experimental study of children, where advances as varied as the evolution of human behaviors that shape markets and institutions to how early life influences shape later life outcomes, have been explored. We first develop a framework for economic preference measurement that provides a lens into how to interpret data from experiments with children. Next, we survey work that provides general empirical insights within our framework and provide a comprehensive summary of experimental methods used with children. Finally, we provide 10 tips for pulling off experiments with children, including factors such as taking into account child competencies, causal identification, and logistical issues related to recruitment and implementation. We envision the experimental study of children as a high-growth research area in the coming decades as social scientists begin to more fully appreciate that children are active participants in markets who (might) respond predictably to economic incentives. (JEL C90, D11, D83, D91, J13, J16, Z13)

1. Introduction

When spanning the research landscape over the past several decades, it is not difficult to find examples of experimental studies where children play a relevant, and in many cases instrumental, role in the advancement of science. Many neuroscientists work

with children as experimental subjects. Developmental and social psychologists rely on children too. Other scientists, including in many fields of medicine, critically depend on children to test their ideas. Indeed, in 1953 Jonas Salk injected his own three sons with a test polio vaccine based on an inactivated poliovirus. This early test paved the way for a nationwide trial that included nearly 2 million

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children. By 1955, the vaccine was widely reported to be safe and effective. While scientists across broad fields of study rely on children to gather data via experimentation, one might wonder—should economists conduct experiments with children too? On several dimensions the answer is unequivocally yes. To organize our thoughts, we classify why economists need children into two silos.

First, in the spirit of medical trials, economists, as well as neurologists, developmental psychologists, sociologists, and the like, must understand children themselves for immediate purposes. One prime example is using experiments to evaluate the impact of different educational interventions (see, e.g., Sadoff 2014; Lavecchia, Liu, and Oreopoulos 2016 for excellent overviews).¹ Indeed, understanding models of human capital formation and early childhood education is an example of understanding children themselves and how best to invest in child development and skill formation. This is because children's choices are modeled as inputs into education production functions that shape their life outcomes (Heckman 2006; List, Samek, and Suskind 2018). Likewise, experiments evaluating health and mental well-being interventions are critical for child outcomes both in the short and long run. More generally, any interventions designed to inform policy prescriptions for children's well-being today fall into this first silo.

Our second silo includes experiments exploring the behaviors and preferences of

children, which we argue provide insights into understanding adults, who are the primary decision makers in most markets. Much like the work in the medical sciences on determining the roots of illness and later life advantages and disadvantages, or the social science work on learned behaviors and language/speech development, economic experiments with children can focus on the decisions and preferences of children that potentially foreshadow adult behavior. As such, experiments with children can be thought of as complementary to recent approaches to understanding factors shaping behavior such as neuroeconomics, which correlates brain activity with decision-making.

Within these two silos, we focus on the emerging literature using experiments with children to understand the determinants of child decision-making, including how economic preferences, information sets, and decision skills affect choices.² Experimental economics has already made headway into understanding the economic preferences of adults and linking them to behaviors and outcomes, ranging from health outcomes such as obesity, to financial outcomes such as credit card debt (e.g., see Falk et al. 2018). Exploring economic preferences, choices, and relevant trade-offs with children is a natural extension. This exploration aligns with the influential work of James Heckman and coauthors in labor economics, who model child skill development with a particular

¹The economics literature in this area is enormous, and spans experimental work using data from the seminal Perry Preschool and Abecedarian experiments (see Heckman et al. 2010; Heckman, Pinto, and Savelyev 2013; summaries are provided in Currie 2001; Duncan and Magnuson 2013; Almond, Currie, and Duque 2018) to work in education research in the 1960s where some children were labeled as high performers and others low performers, when they had actually performed identically on achievement tests (Rosenthal and Jacobson 1968) to work evaluating educational interventions in a developing-country context (see Kremer 2003 for a summary).

²There are several ways to use children in experiments (as Harrison and List (2004) note, such studies fall under the rubric of their definition of “field experiments”). The first is to use children as subjects to evaluate the impact of various interventions on outcomes (i.e., a random controlled trial, or RCT). Since this is a well-traveled area in economics, we only briefly discuss such work so we can focus attention more immediately on the second way, which is to use experiments with children to understand the development of economic preferences and decision-making (this would typically take the form of studying children in the laboratory or in artefactual field experiments).

focus on the importance of “soft” skills (e.g., Cunha and Heckman 2007).³

With respect to the first silo, children are active participants in their human capital production process, and child preferences, beliefs, and information sets are also inputs into household production and consumption. Measurement of such parameters—and mapping them to behaviors—is key to understanding these foundational processes. Further, intervention studies that seek to improve child education, health, or well-being are numerous. Incorporating preference and decision-making experiments into such studies adds value by identifying the role that child preferences play in immediate child behavior (i.e., which interventions will work) and by providing a potentially important variable for heterogeneity, or moderation, analysis (i.e., for whom will interventions work). With respect to the second silo, experiments with child subjects can lend insights into questions about adults. For example, we can measure the variance in preferences associated with differences in family background at an early age, learn the role of experience in shaping preferences; and estimate when traits and preferences emerge and how they evolve through time.

Why is the study of children by economists necessary, given the substantial body of literature on child development from the sister sciences? First, the parameters that economists are interested in may be distinct from those studied by developmental psychologists. For example, in well-known work starting in the 1960s, psychologist Walter Mischel showed that three-to-five-year-old children

who can resist eating a marshmallow in order to receive a larger reward have better educational, labor market, and health outcomes than those who cannot (Shoda, Mischel, and Peake 1990; Schlam et al. 2013). In our own recent work with a group of three-to-five-year-olds, we replicate Mischel’s marshmallow paradigm and conduct time preference experiments that are more standard in the economics literature. We find that these two paradigms measure different constructs, and that time preferences are also predictive of disciplinary referrals in the classroom years later (Castillo et al. 2020).

Second, economists’ training in issues of identification, estimation, and statistical analysis can add value. On the issue of estimation, the theoretical work of Carrol (1962) and other twentieth-century contemporary psychologists of education express learning as a function of the ratio of the time spent on learning to the time needed to learn. Cotton et al. (2020) use a field experiment to identify a structural model to show that these two student-level unobservables—academic efficiency (turning inputs into outputs) and motivation (captured by time preference)—matter for learning across low- and high-performing schools. On the issue of identification, a series of papers have recently emerged suggesting that omitted variables may explain Mischel’s results (Kidd, Palmeri, and Aslin 2013). In this article, we discuss how economics can address such identification issues. Finally, experimental economists have developed a toolkit of incentivized, context-free, elicitation tasks that are portable to the sister sciences that explore complementary behavioral elements, permitting an exploration of economic mediators and moderators.⁴

³The work in labor economics tends to focus on skills as measured by various assessments/tests in education, or with respect to “soft skills,” measurement of character or personality using surveys. Different from that work, tools from experimental economics allow economists to measure economic preferences such as, e.g., time, risk, or social preferences, or to understand how children react in decision problems.

⁴A number of recent papers from psychology have moved from using hypothetical decision tasks to using incentivized games that are more standard in the experimental economics literature. Some of these papers are summarized later in this article.

The remainder of our paper begins with a focus on the broader economics literature that uses children as experimental subjects. These studies span a large academic space, from examining children's rationality, to measuring economic preferences, to an exploration of the variables correlated with children's preferences and behaviors. To place this work in the proper context and aid in its interpretation, we provide a simple conceptual framework. This approach has a dual purpose in that it also permits a deeper understanding of the development of children's economic skills and preferences, and the role of those preferences in shaping behavior.

This exploration in turn informs several fields of economics. For example, experimental economists and economic theorists might find data from children of import to inform preference parameters in economic theories. Labor and education economists might find child data useful to understand the role of children's economic preferences and information sets on behavior that affects their human capital development. Likewise, health economists need information about children's preferences and information sets to inform the design of interventions as well as to study their potentially heterogeneous effects. Finally, researchers studying economics of the household may find value in understanding children themselves as active participants in household bargaining.

We then pivot to a discussion of identification and methodological challenges the experimental researcher faces when using children as subjects. Our goal is to provide practical advice for conducting experiments with children. For example, how can the researcher most efficiently align child competencies with features of the experimental design? In addition, we discuss moving beyond the traditional A/B experimental approach to lend insights into why certain economic choices are taken. This type of

analysis permits a deeper understanding of the behavioral mediators, a necessary step to grasp mechanisms, test theories, and provide efficient policy tools. We next integrate insights from economics and our sister sciences—early education, developmental psychology, and neuroscience—to provide a playbook for how to design and execute field experiments with children. To complement the existing literature, which tends to focus on correlational data, we highlight how experiments can be used to aid in identification and interpretation of data. We also discuss practical issues of consent and ethics, setting up experiments, and selection of incentives. Finally, we conclude by offering ideas on how to scale experiments and best practices for documenting experimental procedures with children. We package this section within a catalog of 10 tips for conducting experiments with children. We conclude with thoughts about where the field is headed and suggestions for future directions that we view as most promising.

2. *The Growing Economics Literature on Experiments with Children*

Experiments with children can provide economists with unique knowledge that cannot be obtained from conducting experiments with adults. This includes studying the development of economic preferences among children in order to understand adults, as well as using economic experiments to understand children themselves to inform models of human capital formation and early childhood education. Economic experiments allow us to measure aspects of children's decision-making such as economic preferences (e.g., risk, time, and social preferences), rationality in decision-making, and information or beliefs that a child possesses. As displayed in figure 1, the past 20 years has seen an enormous increase in working papers and publications that use economic

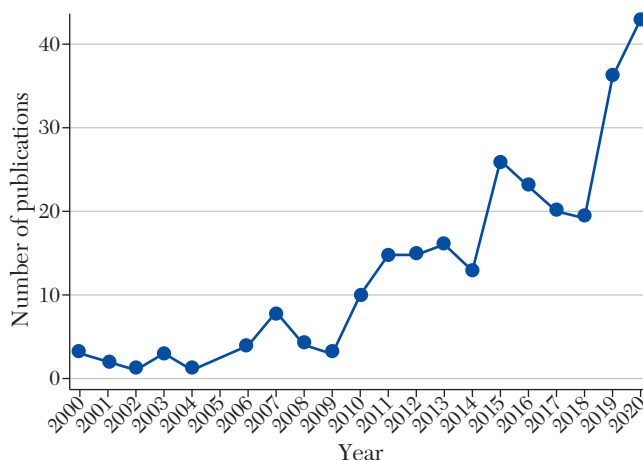


Figure 1. Growth in Economic Experiments with Children

Notes: This figure shows the growth in publications (including working papers posted online) that use experiments with children between 2000 and 2020. The papers included here come from a comprehensive search of the literature by i) searching Google Scholar with keywords “economic experiment” (including searches specific to economic games commonly used in the laboratory) and “children” or “adolescents” and ii) reaching out to the Economic Science Association (ESA) mailing list and to authors of papers identified on Google Scholar to find papers that we may have missed. More details about our search are available in the online appendix.

experiments with children. Given this interest, recent discussions of “alternative subject pools” in experimental economics include children (Fréchette 2016).

Some of the earliest work examining children’s preferences and decision-making in the economics literature comes from William Harbaugh, Kate Krause, Lise Vesterlund, and coauthors. One of the first papers—published in the *American Economic Review*—focuses on investigating whether children are rational decision makers by examining consistency with the general axiom of revealed preference, or GARP (Harbaugh, Krause, and Berry 2001). The authors motivate their work by pointing out that without an understanding of whether children are rational, we cannot know whether standard

economic models apply to them. Harbaugh, Krause, and Berry (2001) find evidence for rationality among children as young as age seven. Harbaugh, Krause, and Berry (2001); Harbaugh, Krause, and Vesterlund (2002); and Harbaugh et al. (2003) expand the exploration to other economic preferences across age groups, including altruism, risk preferences, trust, and behavioral biases. The theme of these papers is to compare decision-making by children to the well-documented decision-making patterns of university students in the laboratory.

The literature grew considerably in the mid-2000s, and ever since, researchers have simplified standard economic games from the laboratory to facilitate understanding among children, including dictator games,

ultimatum games, time and risk preference elicitations, and others. How children behave in these games forms the basis of our understanding of the origins of preferences and decision-making, as well as provides insights for academia and practice.

The Chicago Heights Early Childhood Center (CHECC) is a recent example of a concentrated effort to expand the literature on experiments with children. CHECC is a large-scale field experiment launched in Chicago Heights, Illinois, in 2010. We highlight CHECC because the explicit goals closely follow our two bins on the benefits of conducting experiments with children. The first goal of CHECC is to learn about the optimal investment profile in early childhood to reduce academic achievement gaps. To this end, CHECC randomized over 2,000 children to a control group or to an early childhood program targeting parents or children (see, e.g., Fryer, Levitt, and List 2015; Fryer et al. 2020). Findings suggest that targeting parents has a larger impact on executive functions (Fryer, Levitt, and List 2015) while targeting the schooling input has a larger impact on academic skills (Fryer et al. 2020). Further, findings show that CHECC impacts not only test scores, but brain activity as measured by electroencephalography (EEG). Compared to children in the control group, children assigned to the treatment had greater brain activity related to executive functioning, and this brain activity was predictive of executive functioning skills for up to three years after the intervention (Ye et al. 2021).

The second goal of CHECC is to understand broader issues beyond the intervention effects described above. This includes economic preference development, that is, to understand adults through children. One key to this area of study is understanding how economic preferences, cognitive skills, and noncognitive skills serve as inputs into human capital formation (Castillo et al.

2020). Hence, CHECC also operates as an experimental laboratory and researchers invite CHECC children and parents to participate in periodic economic experiments. CHECC also aims to increase the number of economists studying children. Therefore, another focus is to include researchers outside of the original principal investigators of CHECC to conduct studies. To date, experiments that involve understanding the development of economic preferences with children at CHECC have involved nearly 30 different authors and 11 publications. For these reasons, throughout the article we return to studies involving CHECC to augment the learning from the literature, where appropriate.

3. *Children Can Be Rational*

Before commencing a discussion of the research agenda into the preferences, choices, and behaviors of children, it is important to understand the make-up of this experimental species. For example, it is interesting, but much less so, to study a species that makes decisions haphazardly, or whose preferences are labile and subject to the whims of circumstance. In such cases, no optimization principles may underlie even the most straightforward of choices (see List 2002a).

3.1 *Rationality and Strategic Thinking*

Rationality and strategic thinking form the basis of most economic theories; hence, understanding their development at an early age has attracted the attention of economists. Expanding on earlier work by Harbaugh and coauthors, recent work has continued studying other violations of rationality (Brocas and Carrillo 2018; Castillo, Jordan, and Petrie 2018; Brocas et al. 2019), backward induction (Brosig-Koch, Heinrich, and Helbach 2015), and strategic thinking (Sher, Koenig, and Rustichini

2014; Czermak et al. 2016; Brocas, Carrillo, and Kodaverdian 2017; Brocas et al. 2019). An interesting result is that consistency in decision-making develops asymmetrically across domains. Children first develop consistency in domains that are more familiar to them, for example choosing between toys (Brocas et al. 2019). Because consistency may not be assured when modeling children's decisions, it is important to test for consistency and control for decision error, especially in examining the effects of economic preferences on field behavior (Castillo, Jordan, and Petrie 2018).

Concerning strategic thinking, Brocas and Carrillo (2018) find that children as young as three-to-five years of age can think strategically, though they have limited attention, which makes it difficult for them to consistently act strategically. In Sher, Koenig, and Rustichini (2014), researchers find that by the age of seven, children's ability to act strategically by intentionally misleading is similar to that of adults. Strategic thinking is related to the Theory of Mind (ToM), which is a widely studied construct in developmental psychology that relates to the ability of a child to understand that other people may have different preferences or information states than oneself. The developmental psychology literature suggests that basic ToM develops between the ages of three to five, but that higher-order ToM abilities continue to develop through early adulthood (e.g., see Dumontheil, Apperly, and Blakemore 2010). A recent paper from economics shows that childhood ToM predicts behavior in economic games and is associated with later-life educational and labor market outcomes (Fé, Gill, and Prowse 2022). As such, it is important for economists to understand ToM, and related child traits, not only because ToM is predictive of later life outcomes, but also since models of strategic behavior need to make assumptions about agents' information sets.

3.2 Rationality in Markets

Early field experimental work studying markets includes many child decision makers, providing tests of economic theory among this group of agents (List and Shogren 1998; List 2002a, b, 2003, 2004a; List and Millimet 2008). For example, List (2004a) explores whether predictions from competitive models of equilibrium in bilateral negotiating markets accurately predict behavior. Since standard economic theory relies on two assumptions, utility-maximizing behavior and the Walrasian *tâtonnement* process, relaxing those assumptions remains invaluable. In his lab experiments with Harvard undergraduates, Chamberlin (1948) observes that quantity traded is typically higher and realized prices typically lower than predicted by competitive models of equilibrium. Vernon Smith, an experimental subject in Chamberlin's markets, later refined Chamberlin's work by varying two key aspects of the experimental design in Smith (1962): (i) use of a double-oral auction where an auctioneer took bids and asks; and (ii) use of multiple market periods, permitting agents a chance to learn rules, strategies, and perhaps even valuations in the market (whereas Chamberlin's markets were one shot). Empirical results from Smith's experiments are astonishing—both quantity and price levels are close to competitive predictions. This result is an early example highlighting that Walrasian *tâtonnement* is not a necessary condition in the standard model. By now, hundreds of studies have replicated Smith's original insight.

List (2004a) returns to the Chamberlin institution but moves the analysis from the laboratory environment to a natural setting where the actors had experiences with the institution. Much like Smith's (1962) setup, the market mechanics in these bilateral bargaining markets are not Walrasian. Unlike Smith (1962), however, in these markets

subjects set prices as they please, with no guidance from oral bids and asks (a centralized auctioneer). Thus, List's (2004a) design shifts the task of adaptation from the auctioneer to the market agents, who mill around making bids and asks in a multilateral manner. In doing so, the market structure reformulates the problem of stability of equilibria as a question about the behavior of actual people as opposed to a question about an abstract market. A key result of List's (2004a) study is the tendency for exchange prices to approach the competitive model predictions, especially in symmetric markets (less so in asymmetric markets).

For our purposes, the key data generated in List (2004a) are his experiments with children. That is, the experimental markets in which the only participants are children under the age of 13. These include experienced children—who made more than 25 trades per month in the past two years—and inexperienced children, who made fewer than three trades per month in the past two years. The findings are summarized in figure 2.

The patterns are stark, and provide two insights: first, competitive price theory adequately organizes the data when participants have sufficient market experience.⁵ Second, data from the market sessions with purely inexperienced children are akin to a martingale process, whereby prices follow no apparent trajectory or guidance from theory. Inference from these experiments suggests that even in markets populated entirely by experienced children, there is a tendency toward neoclassical expectations after a few periods. Yet, Chamberlin's (1948) original one-shot insights hold—neoclassical theory does not do well in organizing behavior in

markets populated by agents with little market experience. This exploration showcases the value of experiments with children in that they provide an avenue to study behavioral differences across inexperienced agents and their experienced counterparts in a natural manner. We suspect it would be difficult to find a test group outside of children—or visiting villages without markets—that provides such natural variation. More broadly, it is important to understand children's behavior in such exercises because popular explanations of economic behavior model familial decisions as a part of a household bargaining process in which children are engaged.

The above results are promising since they suggest that with a level of experience children can make rational decisions and think strategically at relatively young ages (or, as in List and Millimet's (2008) tests of GARP, children make a similar level of errors as adults). Researchers wishing to model decision-making by children may consider using standard economic frameworks. However, they should keep in mind other necessary skills that develop concurrently to carry out preferred decisions, such as the ability to reason about others (ToM), attention, and impulse control. Hence, before the ages of 5–7, researchers who model decision-making by children should be concerned about whether it is appropriate to assume standard assumptions of rationality such as transitivity and so on. We discuss the importance, and timing, of child competencies and their implications for design of economic experiments with children in section 9.

4. Conceptual Framework

Having provided evidence of rationality among children, we turn to a conceptual framework that sheds light on how to interpret data on experiments with children. In subsection 4.1, we provide a framework of

⁵A similar result is observed in Brocas and Carrillo (2019a), who replicate List (2004a), finding that 75–82 percent of groups of 5–11-year-old children converge to equilibrium.

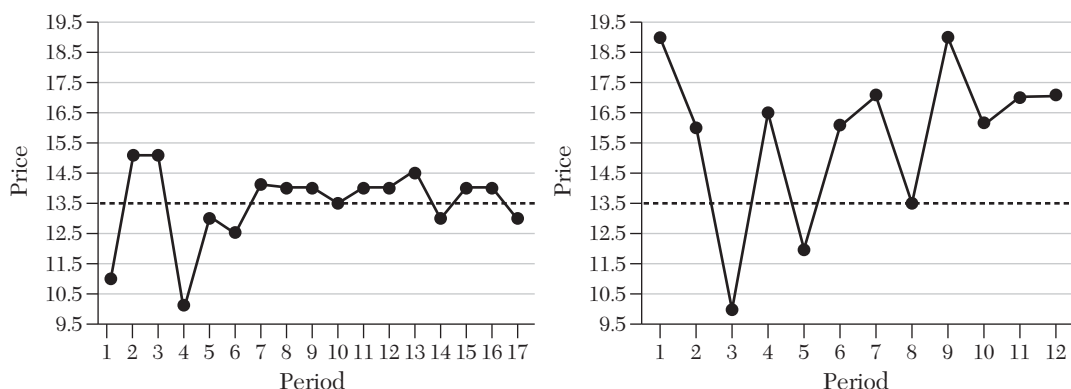


Figure 2. Markets with Children in List (2004a)

Notes: Experienced children (left) and inexperienced children (right).

Sources: This figure is a reproduction of figures 6 and 7 from List (2004a).

human capital development to lend insights into the origins and malleability of skills and economic preferences. In subsection 4.2, we discuss how child skills and economic preferences influence behavior and decision-making.

4.1 Human Capital Development

In our framework, we follow the typical approach, allowing child skills and economic preferences to be affected by investments of parents, schools, and children. Child outcomes are therefore affected by a combination of skills and economic choices, and the latter are partly affected by the child's preferences.

We begin with the assumption that skills and economic preferences result from a nature–nurture (i.e., gene–environment) interaction. This follows the spirit of the genetics literature, which has over the past 25 years moved from a deterministic genetics view (that genetics entirely determine outcomes) toward embracing epigenetics, which posits that gene expression has an

environmental influence (Sternberg 2011; Ijzendoorn, Bakermans-Kranenburg, and Ebstein 2011; Lickliter and Witherington 2017).⁶ Related work in the nascent field of genoconomics suggests that genes explain some—but not all—of the variation in economic preferences and outcomes (Benjamin et al. 2012).⁷

Much of the work that we review in this paper seeks to evaluate the evolution of skills and preferences and to understand what factors affect skill and preference formation. The implicit assumption in this literature is that skills and preferences depend partly on nature and partly on nurture. This means that experiments that seek to understand the intergenerational transmission of economic preferences (discussed in subsection

⁶The idea of a range of potential outcomes available to each individual—even among animals raised in the laboratory—is due to early work by pioneering geneticist Jerry Hirsch (1963).

⁷Some recent examples are studying the heritability of risk preferences (Linnér et al. 2019) and educational attainment (Lee et al. 2018).

6.1) capture both a genetic component and an association with household environment. Some of these papers seek to proxy for genetic background by studying the association of skills and preferences in very young children with mother's skills and household demographic/socioeconomic characteristics. Experiments that seek to understand the impact of shocks or interventions on outcomes (discussed in section 8) are implicitly measuring the impact of environment on gene expression.

In this framework, we discuss the parameters that are relevant to modeling human capital formation. We let M_{it} be a vector of skills and economic preferences for child i at time t . The components of M_{it} could include skills such as cognitive skills, "soft skills," and economic preference parameters representing, for instance, altruism, time discounting, or risk preferences. Each component of M_{it} is the result of a nature–nurture interaction. As such, we can imagine that environmental inputs affect M_{it} , but there are some fixed "bounds" to these effects at the individual level. Initial conditions M_{i0} are presumed to be associated with genetic background.

Skill and economic preference development are affected by inputs of the child, parent, and school, prior skill or economic preferences, norms or aspirational levels, individual characteristics, and bounds on skills. Thus, the state variables for the child's development process in period t are: $\Omega_{it} = (C_{it}^t, P_{it}^t, S_{it}^t, M_{it}^{t-1}, M_i^{t-1}, Z_i)$, where C_{it}^t , P_{it}^t , and S_{it}^t capture the history of child, parent, and school inputs (e.g., time spent learning, time with the child, types of activities with which to engage the child) from the beginning of the process up to period t . M_{it}^{t-1} represents the level of the child's skill and preferences in the prior period, and M_i^{t-1} represents a norm or aspirational level in the prior period. Z_i is individual characteristics, including the bounds on skills.

Cultural norms, peers, and parental preferences are a critical aspect of the human development problem. We introduce a norm or aspirational level, M_i' , to capture the idea that each actor in a child's development may desire for the child's skills or preferences to conform to a group-level norm or a set standard and may incur a cost for deviating from the norm. Such reference dependence is also proposed in Wang et al. (2020), who find that reference-dependent preferences account for a large portion of height differences between children across four villages in Guatemala. M_i' may not be fixed and may evolve endogenously as well as be influenced by exogenous shocks.

A distinction between skills and preferences is that for the former, a higher level of skills is typically assumed to be better. However, for preferences, a normative statement regarding the optimal level of M is in many cases difficult to defend. The optimal level, for example, for risk preferences is less clear, but may be a function of norms, customs, or local rituals.

The production function of skill and preference development is $M_{it} = f(C_{it}, P_{it}, S_{it} | \Omega_{it}, M_i')$, where skills in period t are influenced by inputs from the child, parent, and schools in that period, given the current state and norms or aspirational level. Each actor (parents, schools, and children) makes their investment in child skills, taking the other investments as given, subject to the production function of skills and preferences and a budget constraint. When choosing investments, actors weigh competing interests—for example, parents decide how to allocate their time, money, and effort between investing in their children and their own household consumption (Becker and Tomes 1986).

Parents choose their level of investment to meet their goals for skills and preferences (Bisin and Verdier 2001). Policy makers

implement their preferred minimum level of skills, by setting educational standards for each grade level and holding schools accountable if they are not met. Children—and especially adolescents—make decisions that are in part influenced by decisions of their peers (Brock and Durlauf 2001, 2002; Gavrira and Raphael 2001; List, Momeni, and Zenou 2020; and Durlauf 2004, provide an excellent review; see also papers in subsection 6.3).

When the aims of policy makers and parents do not align, policies can be designed that change the relative price of parental investments in children. For example, the CHECC program we discussed in section 2 shows a causal effect on the amount of time that parents spend teaching their children (Chuan, List, and Samek 2021) as well as causal effects of investment in child skills (Fryer, Levitt, and List 2015). Mayer et al. (2019) find that parenting interventions that incorporate reminders, goal setting, and social rewards are also effective at increasing parental investments. One speculative interpretation that follows from this work is that behavioral aspects—like limited attention—may cause parents to underinvest relative to their optimum.

Cunha and Heckman (2007) proposed a well-known structure for the education production function in which the production of skills and preferences is influenced by skills and inputs from previous periods, and this can reflect self-productivity and dynamic complementarity. Self-productivity means that a skill learned in period $t - 1$ persists and can augment learning of that same skill in period t , and dynamic complementarity means that previously learned skills can bolster investment in new skills in period t . The production function reflects the malleability of realized skills and preferences through aspirations and inputs of the child, parent, and school.

Implications of the Nature–Nurture Interaction.—The literature that we survey in subsection 5.3 evaluates gender differences in competitiveness between men and women (see, e.g., Gneezy, Niederle, and Rustichini 2003). In much of this literature, women display less competitive preferences than men, which is thought to explain in part the corresponding gap in labor market earnings (Gneezy, Niederle, and Rustichini 2003; Niederle and Vesterlund 2011). One might conclude from this evidence that there are fixed “bounds” on competitiveness that are situated to the right (i.e., more competitive) for men compared to women (i.e., less competitive). However, a handful of studies explore behaviors in matrilineal societies and find that, in this context, the gender gap in competitiveness disappears (see, e.g., Gneezy, Leonard, and List 2009; Andersen et al. 2013). Taken together, these findings are consistent with the idea that the bounds for competitiveness overlap for men and women, and we could observe men and women expressing both different and similar M for competitiveness, depending on inputs and environment.⁸

Another implication relates to the impact of interventions on child skills. Heterogeneity in M_{i0} implies that we may also observe heterogeneity in treatment effects. For example, as implied by Cunha and Heckman (2007), higher M_{i0} could make investments in M more productive. Suggestive evidence

⁸There is experimental evidence that given the “right” incentives (i.e., the competitive payment is much higher than the sure payment), women compete as much as men (Petrie and Segal 2017, Ifcher and Zarghamee 2016). Relatedly, in an exploration of gender differences in social preferences, DellaVigna et al. (2013) find that women are more likely to be on the margin of giving, partly because of a less dispersed distribution of altruism. In the same manner, the distributional aspects of M across groups of interest might hold import for the interpretation of gender competition studies. Similar reasoning holds when interpreting empirical results from interventions when setting M' , which is discussed next.

for this is found in the intervention discussed in subsection 2, where the effects of a parenting program were most concentrated among students with high executive functioning skills at baseline (Fryer, Levitt, and List 2015). If bounds on skills exist, the implication is that some children may not be able to reach an aspirational level, M' , if it is set above their bound. For such children, interventions intended to increase achievement in a particular skill to reach a goal may not observe considerable effects in terms of reaching the goal.

We also find evidence for a similar result in the CHECC intervention discussed in section 2—there were no cognitive impacts of the schooling intervention for children who already had high cognitive test scores (Fryer et al. 2020). However, the same intervention could have such an effect on other children if M' is within their bound. Aligned with this, the schooling intervention was highly effective for students with cognitive skills below the median (Fryer et al. 2020). A complementary CHECC study that explores the effects of monthly financial incentives amongst adolescents for meeting an achievement standard based on multiple measures of performance reveals overall modest effects, but has a large and significant impact among students on the threshold of meeting the achievement standard (Levitt, List, and Sadoff 2016). Such results imply that intervention success effects will be heterogeneous just by the nature of the policy maker setting M' . In this manner, a natural set of moderators result directly from how the researcher or policy maker sets M' .

Consider an individual's realized skill or preference level, M_i^* . Two individuals with the same level of realized skill could have very different underlying bounds of potential ability. In other words, improvement in a skill is different from being born into a "good stock" of ability. This insight poses an interesting question with regard to how early

childhood interventions impact later life outcomes. For instance, are long-term outcomes different for children who attained a skill through natural ability or through effort induced by an intervention? This is difficult to tease apart, of course, because the bounds are latent. There is some evidence for "reversion to the mean," in the sense that many early childhood interventions see fade-out of skills learned in the years following a program (Duncan and Magnuson 2013; Almond, Currie, and Duque 2018). This suggests that children who increase their skills through early childhood interventions may need to be scaffolded with additional interventions in the middle childhood years to attain persistent gains, whereas those with higher initial skills may maintain their trajectory without additional scaffolding.

Implications of Properties of the Education Production Function.—According to the production function described in Cunha and Heckman (2007), skills can be complementary in the sense that improvements or changes in skill X could also lead to increases in skill Y . In the literature we survey, there is some evidence that skills move in tandem. For example, studies find a correlation between risk preferences and cognitive abilities (Benjamin, Brown, and Shapiro 2013; Andreoni et al. 2020). Such correlations also have implications for the identification of skills and preferences, which we discuss in subsection 10.4.

Cunha and Heckman's (2007) model suggests that investing in a child early in life leads to the greatest benefits to the skills within M as skills build on one another, and that investments by more parties (e.g., schools, parents) complement one another. In line with these properties, a review of data from interventions targeted at various age groups have shown benefit-cost ratios that decline with the age of the targeted group

(Kautz et al. 2014).⁹ Along these lines, List, Samek, and Suskind (2018) provide a discussion for the importance of investing in children prior to the formal schooling years.¹⁰ Whether actors can shift resources across time and whether markets are complete will impact whether optimal investments are feasible. For example, even if parents are aware that early childhood is a critical period, they may not be able to target investments in that period due to constraints.

The implications for economic preferences have not been studied empirically but follow a similar line of thought. Investments early on are more effective for shifting preferences than investments later, and investments in both schooling and parents are complementary. As discussed in section 7, child preferences early on lead to decisions (e.g., conduct at school) and subsequently affect outcomes (e.g., educational attainment). If preferences shift through inputs, then our work at CHECC in which we assess preferences prior to school investments when children are 3–5 is helpful, since it may come close to providing evidence of the “innate” nature of preferences.

The Role of Beliefs.—An open question is to what extent individuals understand the education production function or the extent to which M is malleable. The concept of mindset in psychology suggests that people differ in their beliefs about whether they can move M . Children who believe they can change their intelligence (growth mindset) achieve more than children who believe that

they cannot do so (fixed mindset) (Dweck 2015). Actors may underinvest in skills if they underestimate the value of investment or the malleability of M . Policies can be designed to counter this issue—for example, Yeager et al. (2019) show that mindset can be changed through interventions.

A recent literature in economics has focused on exploring whether beliefs about the education production function are accurate. This literature generally finds that incorrect beliefs are likely, and further that incorrect beliefs are more common among low socioeconomic status (SES) families. Cunha, Elo, and Culhane (2013) survey low-SES mothers and find that these mothers underestimate the technology of skill formation. The link between SES and beliefs is made clearer in Boneva and Rauh (2018) and List, Pernaudet, and Suskind (2021), who find that perceived returns to parental investments are positively associated with household income.

Parents also may have biased beliefs about their child's skills, M . For example, Kinsler and Pavan (2021) show that parental beliefs about their child's skills are distorted by their local environment—parents of children attending schools with low (high) average skills believe their child is higher (lower) in the overall skill distribution than they are empirically. Such biased beliefs distort investments that parents make in children. Research is needed that explores beliefs about bounds on M or the extent to which M is malleable across subgroups.

Yet, the literature does show that interventions can be designed to affect beliefs. List et al. (2021) find that mothers of babies change their beliefs about the value of parental investments after being exposed to short informational videos combined with home visits. Dizon-Ross (2019) finds that giving parents information about their child's skills affects beliefs and subsequently parental investments. Such interventions are not

⁹However, a newer review did not find this relationship (Rea and Burton 2020). We view this as an invaluable research question going forward that will be better understood through conducting field experiments with children.

¹⁰Investment in the adolescent years should not be discounted, since studies do show impacts of certain investments. For example, a schooling intervention in Joensen et al. (2022) finds that a relatively low-cost ~10 week math class can increase both cognitive and executive functioning skills.

always effective—for example, the home visiting program in Attanasio, Cunha, and Jervis (2019) did not affect parental beliefs. The insights in List et al. (2021) are consistent with this result in the sense that they find “soft touch” interventions do not produce durable belief changes.

4.2 How Economic Preferences Predict Behavior

The literature that we survey also focuses on measuring M (section 5) and evaluating how M correlates with contemporaneous or future field behaviors (section 7). Hence, it is helpful to understand how M enters a child’s utility function and leads to changes in behaviors and outcomes. To do this, we provide a standard formulation of a utility function with preference parameters.

Children are assumed to maximize their utility of consumption over time subject to parental constraints on behavior and a budget constraint that is a function of prices and income, $B_{it} = g(p, m)$, where income and constraints may be determined by their parents. Recall that M_{it} can represent skills and preferences. In this case, M_{it} includes preferences over uncertainty, consumption over time, and preferences towards others. The utility function is $U_{it}(c_{it}|M_{it})$, where $M_{it} = (r, \delta, \gamma)$, r is a risk aversion parameter, δ is a discount rate, and γ is the degree of prosociality towards others.¹¹ Standard assumptions that utility is increasing and quasi-concave over consumption hold. Utility has a curvature to account for risk preferences and a discount factor to account for the weight placed on consumption in

the present versus the future. While utility is presented over consumption, it could more generally represent field behaviors, for example, educational achievement, smoking, et cetera. Choices are affected by these preference parameters in standard ways. More risk averse individuals are less likely to engage in risky behaviors, such as misbehaving in school or smoking. More impatient individuals are more likely to take immediate rewards rather than wait for larger future rewards. Individuals with stronger prosocial inclinations are more generous to others, ceteris paribus. To proxy for these underlying preferences, experiments evaluate child and adolescent decisions in allocations between self and others (subsection 5.1), choices between safe and risky options, and choices between immediate and future rewards (subsection 5.2).

Economists who study field behavior often take preferences as given. However, according to our framework, choices are malleable through investments by children, schools, and parents (subsection 4.1) or constraints placed on a child’s behavior; this prediction is borne out in the empirical data discussed in subsection 8.2. Choices in previous periods may well shape preferences. Children who have fallen behind in school may be impatient and less willing to study to complete coursework that is difficult so that they can graduate. Finally, economic preferences may differ across time because of the development production process and thus yield different outcomes for younger or older children.

Throughout this section, we have made parallel but separate observations about skills and preferences. Human capital development models focus mostly on skills, whereas models that seek to explain behavior focus more on preferences. An interesting question is to what extent one or the other drives outcomes. One recent paper evaluated skills and time preferences of adolescents, finding

¹¹ Social preferences can include separate components for “warm glow,” i.e., utility from the gift itself, and pure altruism, i.e., utility tied to the consumption of others (subsection 5.1). Additional preference parameters that are of interest include present bias, i.e., the overweighing of immediate outcomes (subsection 5.2), ambiguity aversion, i.e., aversion to unknown risks (subsection 5.2), and competitiveness (subsection 5.3).

that academic efficiency (ability to learn) rather than motivation (time preferences) explains why some students fall behind in school and others do not (Cotton et al. 2020). While the Cotton et al. work represents an early empirical contribution attempting to evaluate the core insights of the contemporary psychology of education models of the twentieth century (e.g., Carroll 1962), we trust that future field experiments with children will lend further insights into the underpinnings of adolescent skill formation through the lens of the models of our sister sciences.

5. *Can We Measure Children's Preferences?*

In this section, we provide a discussion of the types of economic preferences of children that have been explored in the literature, including social preferences, risk and time preferences, and competitiveness. As discussed in subsection 4.2, these preferences enter the utility function of the child and influence behaviors and outcomes. A highly recommended companion paper by Sutter, Zoller, and Glätzle-Rützler (2019) provides a comprehensive survey of experimental findings related to economic preferences with respect to age and gender.

5.1 *Social Preferences*

All human interaction is founded on our ability to get along with one another. Humans are unique in their ability to form groups and cooperate with genetic strangers. Social preferences—or preferences toward the welfare of others—are a key component of this feature of humanity. Understanding origins of social preferences is important for our understanding of how institutions form and evolve. For example, the large differences in inequality and redistribution between the United States and Scandinavia are mirrored by differences in inequality acceptance

(but not in preferences toward efficiency) in cross-cultural experiments among adults (Almås, Cappelen, and Tungodden 2020). Indeed, there is a large amount of diversity of social preferences both across and within countries (Falk et al. 2018). Observing how these cross-cultural differences emerge is of interest to the study of political economy, to understand why people contribute time and money to charitable causes or undertake any other economic activity that involves the welfare of others.¹²

To understand social preferences, economists commonly put people in situations where they are given the opportunity to share scarce resources with another person (i.e., a dictator game). At CHECC, in a series of experiments, we gave young children six stickers and asked how many they would like to share with another anonymous child (Castillo et al. 2020). Data collected in the experiments can be found in List, Petrie, and Samek (2022). Figure 3 summarizes these data. Some clear patterns emerge. First, similar to dictator games with adults, we observe a large concentration of sharing at the 50/50 “fair” norm, and children rarely share more than half of their endowment (panel A). We also observe a concentration in zero sharing—giving none of the endowment to the other person. Yet, there remains marked heterogeneity in allocation decisions. Second, even at young ages, children somewhat increase their sharing behavior as they grow older, from ages 4 through 6.5 years old, both in terms of the average amount of endowment shared and the likelihood of sharing at the 50/50 norm (shown in panel B).

A similar, and even more pronounced trend of increased sharing with age has been

¹²Arrow (1972) perhaps put it best when noting: “virtually every commercial transaction has within itself an element of trust . . . it can be plausibly argued that much of the economic backwardness in the world can be explained by the lack of mutual confidence.”

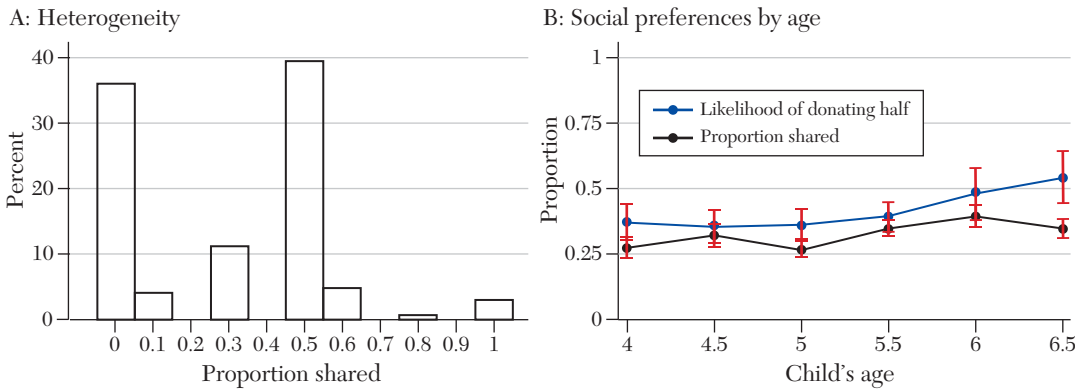


Figure 3. Heterogeneity in Social Preferences and Social Preferences by Age

Notes: This figure uses data from CHECC reported on in Castillo et al. (2020) to plot a histogram showing heterogeneity in social preferences (panel A) and preferences by child age, from 4 to 6.6 years old (panel B). Data is depicted as proportion of the reward shared (out of 6). Here, we only use “Wave 2012” data because this is the data for which we have the largest age range. Panel B also includes the proportion who share exactly half of their endowment.

documented in a number of studies over the past decade (e.g., Fehr, Bernhard, and Rockenbach 2008; Angerer, Glätzle-Rützler et al. 2015). These studies use dictator games similar to the work at CHECC, as well as allocation decisions with restricted choice sets that allow for direct comparisons of, for example, fairness and efficiency preferences. Studies also have found that trust, reciprocity, and cooperation increase with age (Harbaugh et al. 2003; Sutter and Kocher 2007; Evans, Athenstaedt, and Krueger 2013). To study trust, reciprocity, and cooperation, researchers typically use simplified variants of commonly used paradigms in laboratory experiments, for instance, ultimatum, prisoner’s dilemma, and public goods games (Ledyard 1995). In a field study to explore social preferences of individuals across collector conventions (in public goods games), a television game show (in prisoner dilemma games), and a university fundraising drive (charitable gifts), List (2004b) finds that in every setting there is a strong

relationship between age and other-regarding preferences—in every case the older subjects act more pro-socially.

Rather than showing how behaviors in games change with age, another research approach is to design experiments that allow us to distinguish between theories. For example, List and Samek (2013) study whether dictator game decisions at an early age were due to warm glow or pure altruism, finding evidence for pure altruism but not warm glow. Fehr, Glätzle-Rützler, and Sutter (2013) explore the evolution of egalitarian, altruistic, and spiteful preferences, finding that altruism became more prominent with age, but that spitefulness and egalitarianism became less frequent with age. Bereby-Meyer and Fiks (2013) find that negative reciprocity increases with age. There are also some instances in which theories apply equally well to children as they do to adults. For example, the theory of moral cost described in Cox et al. (2016), whereby people make decisions based partly on the

action set available to them, predict the behavior of children as young as three as well as the behavior of adult subjects.

Related work also uses social preference experiments with children to understand preferences toward in-group or out-group members, or discrimination. For instance, List, List, and Samek (2017) use dictator games that vary whether the recipient appears White or Black and find that three-five-year-old White and Hispanic children send more to the Black recipient than to the White recipient—a pattern that is reversed in studies of adults (Riach and Rich 2002). Related work finds that language discrimination in a bilingual city in Italy increases with the child's age suggesting that perhaps preferences regarding minorities develop with age (Angerer et al. 2016).

An open question revolves around what triggers social preferences to manifest in children. A potential explanation in the case of fairness preferences is that children become more concerned about fairness after they have developed the math skills that allow them to create and comprehend an equal division of the endowment (Chernyak et al. 2016). This underscores the importance of considering cognitive skills that develop concurrently when measuring economic preference development. Such within-period complementarities of skills are captured in our conceptual framework of skill formation. Other explanations about why social preferences increase with age include socialization (Fehr, Glätzle-Rützler, and Sutter 2013) and the ability to take into account the preferences and viewpoints of others (Bereby-Meyer and Fiks 2013).

5.2 Time and Risk Preferences

Time and risk preferences play an important role in a wide range of decision-making contexts, which are of interest to economists in fields that span from household finance to health to education. To measure time and

risk preferences, economists usually put people in situations where they make decisions about receiving rewards sooner versus later, and about receiving sure rewards versus being entered into a lottery for rewards, respectively.

Research shows that time preferences, and especially present-biased preferences, are related to participation in financial education programs and credit card borrowing among adults (Meier and Sprenger 2010, 2013). These preferences are also related to health outcomes—for example, impatience is correlated with higher body mass index (BMI) (Ikeda, Kang, and Ohtake 2010; Sutter et al. 2013; Golsteyn, Grönqvist, and Lindahl 2014; Weller et al. 2008) and more smoking, drinking, and drug abuse behaviors (Bradford et al. 2017, Chabris et al. 2008). Risk tolerance is also linked to smoking and drinking (Anderson and Mellor 2008). Finally, time preferences are linked to misbehavior, educational attainment, and lifetime income (Castillo et al. 2011; Castillo, Jordan, and Petrie 2019; Golsteyn, Grönqvist, and Lindahl 2014).

Understanding time and risk preference development is important for several reasons. First, we want to understand the origins of such preferences and how they are shaped by parents and schooling. Second, we want to understand time and risk preferences of children to inform policy interventions. For example, most studies find that patience increases with age (see Sutter, Zoller, and Glätzle-Rützler 2019 for an excellent overview). This suggests that interventions targeting children should focus more on the present than on the future—that is, when giving incentives for reaching education or health goals, these incentives should be distributed as soon as possible (Levitt et al. 2016). Alternatively, adults may be more willing to accept delays in receipt of their incentives. As another example, researchers worry about the apparent greater risk

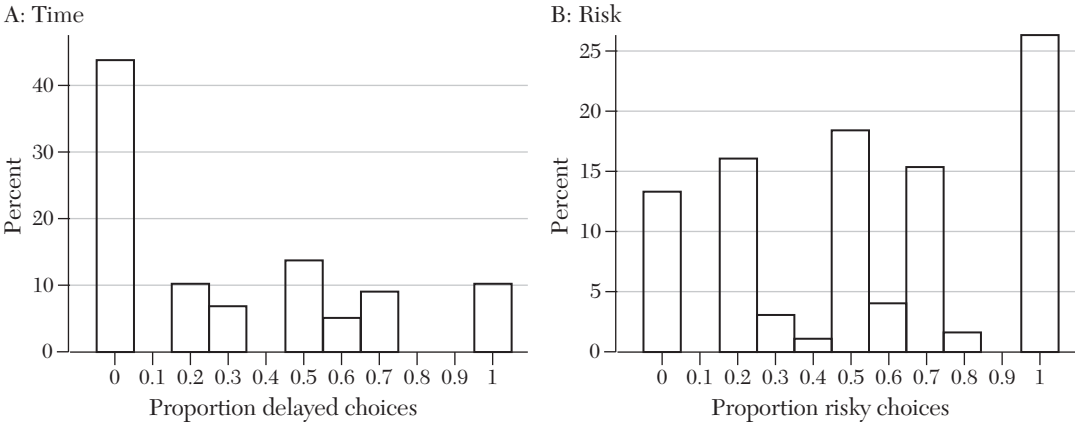


Figure 4. Heterogeneity in Time and Risk Preferences of Young Children

Notes: This figure uses data from CHECC reported on in Castillo et al. (2020) to plot a histogram showing heterogeneity in time preferences (panel A) and risk preferences (panel B). Data is depicted as the proportion of delayed choices in a menu of earlier/later decisions and proportion of risky choices in a menu of safe/risky decisions.

tolerance of adolescents that can lead to detrimental behaviors like smoking and drinking, yet experiments show us that we should be most worried about their tolerance toward ambiguity, that is, toward unknown risks (Tymula et al. 2012).

Using the CHECC data, Castillo et al. (2020) find that time preferences measured at three–five years of age are associated with disciplinary referrals years later in elementary school. Importantly, these preferences are distinct from cognitive skills (e.g., reading, writing, math) and executive functions (e.g., inhibitory control and working memory) in that they have an independent role in predicting disciplinary referrals. Hence, Castillo et al. (2020) refer to economic preferences as a key factor previously missing in our understanding of human capital formation.

Figure 4 shows the raw data from time and risk preference experiments we conducted

at CHECC when children were three–five years old. In time preference experiments (panel A), we gave children choices between smaller, earlier candy rewards and larger, later rewards.¹³ In risk preference experiments (panel B), we gave children opportunities to choose between smaller, certain rewards or different probabilities of larger rewards.

In figure 4, we immediately observe heterogeneities in economic preferences, even at these young ages. This lends empirical support for the conceptual framework in section 4. There is a range of behaviors manifested by the children. How these heterogeneities arise, and how malleable

¹³Children also completed the “marshmallow” task (Mischel, Shoda, and Rodriguez 1989). In Castillo et al. (2020), we show that the time preference task and the marshmallow task are distinct drivers of behavior and only time preferences were predictive of disciplinary referrals.

they are, remain open questions for future research.

5.3 *The Competitiveness Gender Gap*

A concern in the field of labor economics is that in many developed countries women are pursuing higher education at similar, or higher, rates than men, yet women continue to have relatively lower earnings and are less likely to hold executive positions in firms (Bertrand 2009; Blau, Ferber, and Winkler 2010; Bertrand and Hallock 2001; Wolfers 2006). Gender differences in competitiveness have been proposed as one reason for this gap, and a large literature has used experiments to show that men and women have heterogeneous preferences over wage contracts, with men preferring relative wage incentives more than women (Gneezy, Niederle, and Rustichini 2003; Niederle and Vesterlund 2007; Gneezy, Leonard, and List 2009; Flory, Leibbrandt, and List 2015—Niederle and Vesterlund 2011 provides a review).

For their part, experiments with children have begun to inform us when this gap develops and what we can do about it. In an early paper on this topic, Gneezy and Rustichini (2004) asked Israeli children to run a short distance either alone or in competition. The authors find that competition improved the running time of boys, but not of girls. The work that followed has mostly preferred a choice-based measure of competitiveness preference rather than a performance-based measure. Most of this work followed Niederle and Vesterlund (2007) by asking children to perform a task under a piece rate and under a tournament, and then eliciting a choice of piece rate or tournament to assess willingness to compete.

An overview of the findings in this literature is that competitiveness increases as children get older, and that girls are generally less competitive than boys (for an

example, see Sutter and Glätzle-Rutzler 2015).¹⁴ However, there are some important additional takeaways. First, culture seems to have a large influence. The gender gap in the running task is not observed among Swedish children or Armenian children (Dreber, von Essen, and Ranehill 2011; Khachatryan et al. 2015), in contrast to Israeli children (Gneezy and Rustichini 2004). Nor is there a gender gap in ball throwing in a matrilineal society in India (Andersen et al. 2013). Girls from same-sex schools are as competitive as boys (Booth and Nolen 2012). Second, task differences also seem to be important. For example, the gender gap in competitiveness is not observed in the “female” sports of skipping rope and dancing (Dreber, von Essen, and Ranehill 2014; Khachatryan et al. 2015).

The literature suggests that culture, norms, and social learning play critical roles in the development of competitiveness preferences. Cultural norms of behavior may impose rigidities on the expression of skills so that patterns of gender differences in competitive preferences seemingly reverse (see Andersen et al. 2013, as an example). Culture and norms may also play a role in time, risk, and social preferences, where there is as much diversity across countries as within (Falk et al. 2018). This highlights the import of task selection (and potentially needing many types of tasks) to elicit a competitiveness trait.

¹⁴Yet, the work of Flory et al. (2018) shows that while research on competitiveness has emphasized sex as a physiological determinant, their evidence shows that women's preferences over competition change with age such that the gender gap, while large for young adults, disappears in older populations because older women are much more competitive. This result suggests that a simple gender-based view of competitiveness is misleading, and that age seems just as important as sex. Future field experiments with children in this space can bear valuable fruit, as early life events will surely be meaningful.

6. What Variables Correlate with Children's Preferences?

A key to understanding the causes of children's behavior is having a firm grasp of their preferences, not only the origination of such preferences but also the correlates. Children can be influenced by a myriad of factors, from their parents to their neighbors to their environment. This section explores correlates with children's behaviors from a broad variety of empirical approaches but focuses on studies using field experiments, as defined in Harrison and List (2004).

6.1 Children and Their Parents

Children spend a disproportionate amount of time with their parents, especially when they are young. As described in our model in section 4, the influence that parents have on their children could affect their skills and economic preferences for their patience, willingness to take risks, competitive attitudes, and prosocial behavior. The literature on improving the outcomes of disadvantaged children has started to focus on parental investments (see List, Samek, and Suskind 2018, for an overview). Fryer, Levitt, and List (2015) use the CHECC program to explore the impact of intervening with parents on improving child noncognitive skills.

Researchers hypothesize that parenting shapes economic preferences (Bisin and Verdier 2001, Lundberg et al. 2009, Doepke and Zilibotti 2017, Del Boca et al. 2019). This could occur as the child grows and decisions shift from being made by the parent to being made by the child. During this time, parents transmit their preferences to the child, whether the child might agree with their parent or not (possibly through parental investments in the child, see our model in section 4). Understanding if and how this occurs requires an empirical evaluation of children and their parents.

Certain evidence shows that parents and their adult children share similarities in economic preferences (Dohmen et al. 2012; Kimball, Sahm, and Shapiro 2009). However, research with parents and children finds mixed evidence of a correlation for patience (Bettinger and Slonim 2007, Kosse and Pfeiffer 2012, Andreoni et al. 2019, Samek et al. 2021) and no evidence for a correlation in altruism (Ben-Ner et al. 2017, Bettinger and Slonim 2006). There does seem to be a correlation for risk preferences between children and parents (Alan et al. 2017; Chowdhury, Sutter, and Zimmermann 2018; Samek et al. 2021).¹⁵ Further, Khadjavi and Nicklisch (2018) find that parents' ambitions for their children are correlated with child competitiveness. We interpret the literature as suggestive evidence that parents do indeed transmit their preferences to their children. Furthermore, the finding of a more persistent correlation among adult children than among young children suggests that many years of parental socialization are required for preferences to "stick." An alternative explanation is that the correlation exists even at young ages, but that preferences are difficult to measure—or are noisier—in early childhood.

A related area of study is how children affect decision-making in the household and how that decision-making affects a child's choices. A child's preferences interact with her parent's preferences, and this can affect many aspects of behavior within and outside of a household, such as human capital development of children, products purchased in the household, and constraints parents might place on a child's time. The interaction between parent and child within the household is typically modeled as a noncooperative game and preferences are inferred from survey data or parameterized from auxiliary

¹⁵ Alan et al. (2017) study mothers and the correlation is only present for daughters.

experimental data sets (Lundberg, Romich, and Tsang 2009; Del Boca et al. 2019).

Examining the interaction of children and parents in decision-making, Houser et al. (2015), using children from CHECC, experimentally examine the effect of having a child present when parents make a decision whether to cheat. They find that in the presence of girls—but not of boys—parents are less likely to cheat when their child can observe their behavior, especially when the cheating benefits the child. In another CHECC study, Ben-Ner et al. (2017) examine the impact of children on donation decisions of their parents. In this study, parents are asked to make a donation decision that in some cases is observed by their child. The authors find that some subsets of parents (fathers, and parents of generous children) are more likely to model generous behavior when they know that their child will see their decision. These results suggest that parents do make decisions aimed at influencing children's preferences, as we predict in our model in section 4—for example, modeling honest or generous preferences in a potential attempt to encourage their children to adopt such preferences.

6.2 *Children and Their Socioeconomic Status*

Inequality is an important societal issue that is of interest in all fields of economics. What explains inequality, and how to reduce inequality, are open questions. Dynarski et al. (2021) conduct a field experiment to evaluate the impact of information and promise of financial assistance upon admission on encouraging low-income, high-achieving students to enroll at the University of Michigan. The number of students who apply and enroll doubles, suggesting such an approach could have strong impacts on closing income gaps through college choice.

One possibility is that inequalities are persistent due to differences in economic

preferences between different populations. Indeed, experiments with children show associations of SES with economic preferences. Research shows that children and adolescents from low-SES backgrounds tend to be less patient and more risk seeking than higher-income counterparts (Eckel et al. 2012; Deckers et al. 2015; Falk and Kosse 2016). With one exception (Almås et al. 2017), most studies also find that lower SES children are less altruistic/egalitarian (Benenson, Pascoe, and Radmore 2007; Bauer, Chytilová, and Pertold-Gebicka 2014; Deckers et al. 2015; Falk and Kosse 2016; Kosse et al. 2020).

The findings are relevant since exhibiting more impatience as a young child or adolescent could have compounding effects on educational choices—for example, misbehavior in school or graduating from high school (Castillo et al. 2011; Castillo, Jordan, and Petrie 2019; Castillo et al. 2020). Adolescents who are more tolerant to risk may take up drugs or other risky behaviors, which could also impact their future outcomes (Rao et al. 2011). It is less obvious how other preferences—for instance, social preferences—affect inequality. However, as children become adults and start making decisions that affect institutions, more egalitarian preferences in society can lead to different policy decisions (e.g., with regards to taxation and redistribution) (Cappelen et al. 2020).

An important caveat is that we should not draw causal inference from the studies described above. It is not clear whether differences in SES cause the differences in preferences, or whether omitted variables drive these differences. Omitted variables could include differences in access to resources, differences in parenting, or differences in cognitive abilities. Incorporating these variables into the analysis will help to limit (but not fully solve) this problem. Another open question is whether differences in

preferences by SES are present at birth, or whether interactions with the environment result in these differences. There is some evidence for the latter in a study by Rao (2019)—the economic preferences of adolescents from higher-income backgrounds are affected by random assignment to classrooms with adolescents from lower-income backgrounds.

6.3 *Children and Their Peers*

In the framework developed in section 4, we discuss a group-level norm of M' that incentivizes the child to conform to the norm. To this end, related experimental literature evaluates the impact of peers and neighborhoods. In List et al. (2020), the authors take advantage of the CHECC field experiment to explore the social side of human capital formation and skill development. While most of behavioral economics is founded on psychological concepts, List et al. (2020) leverage insights from sociology (for example, Coleman 1988) to examine neighbor and peer effects on human capital formation of three–five-year-olds. They are able to make causal statements on neighbor effects because CHECC randomized treatment allocation across space and time, effectively creating a checkerboard panel data set to exploit. List, Momeni, and Zenou (2020) document large, localized spillover effects on control children who live near treated children. For our purposes, perhaps the most compelling evidence that speaks to our theoretical framework is the underlying economic mediator (mechanisms) at work. They find that the spillover effect on executive functioning scores operates through the child's social network while parental investment is an important channel through which cognitive spillover effects operate. Both results highlight that human capital accumulation is fundamentally a social activity, even at very young ages.

Peer effects are also explored in Rao (2019), who examines the effects of a change in school policy in India whereby children from poor families are placed in classrooms with rich families. Rao (2019) collects administrative and experimental data on affected children and finds that children from rich families are more prosocial after having been exposed to children from poor families. By contrast, there are limited impacts of such exposure on academic achievement.

Another approach to understand the impact of peers is to conduct manipulations in which child decisions are either made public or private. This is most commonly done to study social preferences. In work that uses this approach, children as young as five are more generous when the recipient is aware of their donation options (Leimgruber et al. 2012). Similar results are found in Houser, Montinari, and Piovesan (2012) for children aged nine and above when their decisions are made in public environments, and in Chen et al. (2016) for children aged eight when their choices are displayed to their peers. Children also tend to be more generous toward in-group than out-group members, see for example Bindra, Rützler, and Lergetporer (2020) for a study with children ages 3–6, Bauer et al. (2014) for ages 7–18, Sparks, Schinkel, and Moore (2017) for ages 4–6, and Zhu, Guan, and Li (2015) for ages 2.5–6.5.

Related work finds some evidence for peer effects on other economic preferences of children, but this work is limited. Eckel et al. (2012) find that a higher percentage of low-income peers in a school is correlated with greater risk aversion among adolescents. However, this paper does not seek to identify whether this was a causal effect. Charness et al. (2019) find that prompting children to think about peers improves their performance on a ToM task.

7. *Do Preferences Correlate with Field Behavior?*

A large literature evaluates the association of adults' economic preferences with field behaviors. This work finds that economic preferences are associated with field behaviors in ways that are consistent with the framework in subsection 4.2. For example, increased patience is associated with savings behavior, lower credit card debt, and improved health outcomes, which are also associated with what we view as positive life outcomes (Chapman and Coups 1999; Picone, Sloan, and Taylor 2004; Weller et al. 2008; Bradford 2010; Meier and Sprenger 2010; Leonard et al. 2013). Research with children and adolescents finds similar associations. These results are important since they suggest that economists are measuring a construct that also has field relevance among children. Importantly, however, observing preferences and behavior at the same time does not speak to causal effects as preferences and behavior may co-evolve.

The literature does teach us that time and risk preferences of children and adolescents are correlated with smoking, drinking, body mass index (BMI), savings, and conduct at school some of the time (e.g., Sutter et al. 2013). Focusing on a small sample of truant adolescents, risk and time preferences are not found to be correlated with school attendance (Antrobus et al. 2017). Samek et al. (2021) find an association of time preferences with the BMIs of adolescents when time preferences are measured using a stated-preference survey, but not when preferences are measured using an incentivized experimental elicitation task. In another paper, competitive preferences are correlated with adolescent choices of academic track in school (Buser, Niederle, and Oosterbeek 2014).

Increasing the distance in time between the measurement of preferences and field

behavior allows for an examination of the significance of early preferences on future outcomes. Issues of causality remain relevant—for example, other coevolving omitted variables could drive the associations, increasing causal density. This problem can be mitigated somewhat by including a large number of covariates as control variables. Finding that measured preferences earlier in life are correlated with longer-run behavior confirms the importance of understanding children to understand adults, yet in the end the results remain correlational.

In a nutshell, however, the literature does show that time and risk preferences measured in adolescence are correlated with educational and labor market outcomes later in life. More impatient and risk-taking adolescents, as measured with incentivized experiments, are more likely to misbehave in school (Castillo et al. 2011; Castillo, Jordan, and Petrie 2018) and are less likely to graduate from high school five years later (Castillo, Jordan, and Petrie 2019), even controlling for demographics, cognitive ability, and school effects. Other studies using unincentivized measures find a relationship between present-biasedness, self-control, and impulsivity in children, adolescents, and young adults and labor market and health outcomes (DellaVigna and Paserman 2005; Moffitt et al. 2011). Golsteyn, Grönqvist, and Lindahl (2014) use a hypothetical time preference measure with teenagers and find a relationship with labor market outcomes 20 years later. Cadena and Keys (2015) use a measure of impatience from an interviewer report of whether the young adult is “impatient or restless” during the interview and find that it is related to school drop-out and labor market outcomes later in life.

Examining behaviors at even younger ages, we find that there is evidence for preferences playing a role in life outcomes as well. Mischel, Shoda, and Rodriguez (1989) show that a child's ability to resist eating a

marshmallow at age four predicts social and cognitive competency during adolescence (see also Mischel, Shoda, and Peake 1988; Shoda, Mischel, and Peake 1990). However, some scholars challenge the robustness of these findings with additional data (Watts, Duncan, and Quan 2018). In particular, the marshmallow paradigm may not be a clean measure of the ability to delay, since manipulating the uncertainty in the environment also has large impacts on delay decisions (Kidd, Palermi, and Aslin 2013). Further, Benjamin et al. (2020) collect data from the original Mischel and Ebbsen (1970) sample 40 years later and find no effect of the duration of time waited on a variety of life outcomes, but they do find an impact of an index of self-regulation measured at various later ages on adult outcomes. This study suggests that the marshmallow paradigm could also be picking up aspects of the child's environment or is less informative for later life outcomes than advertised in the literature and popular press. It also points again to the importance of considering omitted variables as discussed in subsection 6.2.

Exploring behaviors in CHECC children, we find that child time preferences as measured by incentivized experiments at three–five years of age explain disciplinary referrals in elementary school up to six years later, even when controlling for cognitive abilities and executive functions (Castillo et al. 2020). Importantly, we find that economic preferences are a third, distinct factor shaping life outcomes (in addition to cognitive skills and executive functions).

8. *Identification of Preference Development*

Discovering causal relationships is invaluable since they represent the key to theory testing, the basis of fact, and are vital for informed policy prescriptions. For example, one important exercise is to identify

the impact of the inputs—child, parent, and school—on human capital formation, as discussed in section 4. Using naturally occurring data (in many cases “natural experiments”) and invoking various assumptions represents the standard approach to providing such causal estimates in economics. However, the economics community has turned increasingly to using random assignment or field experiments with well-specified models.¹⁶ For some policy questions of interest, this approach can be challenging to implement or potentially unethical or ill-advised. For example, to test the effects of domestic violence or low SES on a child's economic preferences, we would not want to force individuals into disadvantaged circumstances. Instead, interventions should have the intention of delivering positive change. For example, random assignment to income transfers to increase the household budget would be an appropriate way to experimentally measure the effects of a change in income. Or, an intervention to reduce household violence could be used to evaluate the effect of violence on child skills and preferences.

There is a large and growing literature on the effects of early childhood or in utero environmental effects on child cognitive skills and misbehavior (Almond and Currie 2011). Yet research on the effects of interventions, natural experiments, environmental conditions, or negative shocks on economic preferences is still in its infancy. Our model in section 4 suggests that it is also important to understand these effects, since we consider preferences malleable. Interventions aimed at improving household financial well-being—such as cash transfers or interventions aimed at improving child cognitive skills through educational programming—may have unintended (positive or negative) consequences

¹⁶Fryer (2017) provides a review of randomized field experiments in human capital formation.

on economic preferences.¹⁷ In turn, preferences shaped in childhood may affect individual outcomes in adulthood, as well as voting and decision-making that potentially impacts institutions more broadly.

When measuring contemporaneous environmental factors and economic preferences without relying on experimentation, there can be a problem of reverse causality. A good example of a reverse causality problem is in the nutrition field, which reports that restrictive parent feeding practices are positively correlated with childhood obesity (Shloim et al. 2015; Boswell, Byrne, and Davies 2019). Is it the case that obesity is caused by restrictive feeding practices, or that obesity causes parents to restrict feeding? Outside of an experimental evaluation, one solution is separating the evaluation in time, as is done in several papers in section 7—for instance, measuring feeding practices at $t = 0$ and returning a year later to measure BMI at $t = 1$. This separation in time must still address the omitted variables problem discussed in subsection 6.2, since other factors could be driving the correlational effects observed. While one could strive to incorporate all manner of observable characteristics as controls, other unobserved drivers of behavior are still possible. This points again to the benefits of randomization for identification.

8.1 *Exploiting Natural Experiments to Understand Preferences*

Economists have been exploiting natural experiments for decades. A common example of a natural experiment uses changes in compulsory schooling rules to understand the returns to education (e.g., Pischke and

von Wachter 2008; Bhuller, Mogstad, and Salvanes 2017). Natural disasters, wars, bombings, and civil conflict are other exogenous events that can have a profound effect on outcomes and preferences. Bauer et al. (2014) show that children exposed to war become more egalitarian toward their in-group. Castillo, Petrie and Torero (2019) find that being exposed to political violence while in utero produced a higher willingness to take risks 20–30 years later as an adult, and Castillo (2020) finds children exposed to domestic violence are more risk averse at age 12. Callen et al. (2014) examine how psychological priming interacts with exposure to violence zero–eight years earlier on risk preferences. Adults exposed to violence up to eight years earlier and primed with fear were more likely to prefer certain outcomes relative to uncertain outcomes.

Income shocks can also be used as natural experiments. Carrillo (2020) examines the variation in coffee cultivation in Colombia during children's schooling years and finds that cohorts that faced higher returns to coffee-related work in childhood completed less schooling and had lower adult earnings. He suggests that these findings are consistent with the possibility that students ignore or heavily discount the future when faced with immediate income gains.

Changes in school policies and assignment to classrooms and schools may provide interactions with new groups of individuals. We discussed studies that take advantage of these new interactions to study peer effects in subsection 6.3 (List, Momeni, and Zenou 2020; Rao 2019).

8.2 *Interventions and Effects on Preferences*

The literature on interventions to understand the stability and malleability of preferences is of two types. The first type develops a program that specifically targets a particular economic preference and evaluates the impact of the program on that preference.

¹⁷There is a large literature on the effects of conditional cash transfer or support programs on a variety of life outcomes (e.g., Hoynes, Schanzenbach, and Almond 2016 on the Supplemental Nutrition Assistance Program in the United States and Attanasio, Meghir, and Santiago 2012 and Parker and Todd 2017 on PROGRESA in Mexico)

For example, Alan and Ertac (2018) find that a curriculum targeting forward-looking behavior and imagining a future self increases patience in incentivized intertemporal choice experiments. Lührmann, Serra-Garcia, and Winter (2018) find that a financial literacy program increases time consistency. Alan, Boneva, and Ertac (2019) find that an educational intervention teaching grit increases persistence in a real effort task, improves math test scores, and reduces the gender gap in competitiveness (Alan and Ertac 2019). Based on the framework in section 4, such interventions might work by changing the inputs of the schools, S .

The second strand of this literature evaluates the impact of interventions with alternative aims—for example, programs aimed at reducing the academic achievement gap—on economic preferences. For example, the CHECC program was designed to impact educational attainment through early interventions with children and parents. This included a Parent Academy program that targeted parents (through classes on how to teach to children at home) and a preschool program that targeted children (by providing a free, full-day high-quality preschool). Because these programs affect inputs P and S , respectively, they might also affect economic preferences. For example, the Parent Academy program might affect children through changing how parents interact with their children or how they model behavior, while the preschool program might affect children through exposing them to interaction with peers and teachers.

Cappelen et al. (2020) report causal evidence of the impact of these programs on social preferences. They find that the CHECC Parent Academy increases the relative weight children place on efficiency versus fairness, while the preschool makes children more egalitarian in their fairness views, relative to the control group. To

explain these results, Cappelen et al. (2020) suggest that parents modeled more efficient allocations in the Parent Academy treatment, and that the preschool exposed children to sharing rules. A related study by Kosse et al. (2020) finds that a mentoring program aimed at enriching the social environment for children ages seven–nine increases prosociality by 27 percent of a standard deviation relative to the control group. Similarly, Bettinger and Slonim (2006) examine the effect of school vouchers on prosociality of children (ages 8–16 years old). They find that children in families that received an educational voucher are more generous when given the opportunity to make donations to a charity but are no more generous when given the opportunity to share with a peer.

There is little evidence on the long-run effects of interventions on preferences. It is important to collect long-run data because such effects may emerge later in life. For example, displacement effects of moving out of public housing in Chicago due to demolition show minimal short-run effects (Jacob 2004), but have large positive effects on long-run outcomes, such as employment and earnings (Chyn 2018). Research shows that the effects of early childhood interventions tend to fade out in the intermediate term, yet the Perry Preschool project has positive impacts on treated children in adulthood, and further, this program has intergenerational effects on the children of treated children (Heckman and Karapakula 2019).

Studying moderators, or the heterogeneous impact of an intervention, benefits from understanding the children themselves. For example, Fryer et al. (2020) show that the CHECC preschool is most beneficial for children with below-median cognitive and executive functioning skills. A related area is to explore how economic preferences interact with an intervention. One example of such a study is by Datar, Nicosia, and Samek

(2022), who find that adolescent time preferences mediate the impact of environment on obesity. There are additional open questions related to interaction effects. For example, what determines parental investments in a child and how might that interact with child behaviors or preferences? Also, how much does the mapping (or returns) of parental investment vary across children?

9. *Child Competencies and Experimental Design*

The disciplines of developmental psychology, education, and neuroscience provide insights into the child development process, but the design features of economic experiments are unique and require further interpretation. The rapid increase in experiments with children is promising, but it is important to step back to consider how best to design experiments for every age range. This can reduce noise in the data and make comparisons across experiments more useful. While we cannot say what would constitute the best practice yet, we make steps in this direction in several ways.

First, we discuss the development of child competencies taken from the sister sciences—developmental psychology, education, and neuroscience—that are important to consider when designing economics experiments. Second, we summarize the methods used in experiments with children of different ages to date, including organizing our thoughts around what we think are the key design and protocol considerations and synthesizing the above to provide suggestions for how competencies can be taken into account when designing experiments. To assist in this task, we must integrate an understanding of when competencies develop (from the developmental psychology and cognitive neuroscience disciplines) with an understanding of what educational

tools are effective for explaining concepts to children at different ages (from the education discipline).

9.1 *Child Competencies to Consider in Design*

Two core child competencies that we must consider for the design of experiments are academic skills (e.g., literacy, numeracy) and executive functions (working memory, inhibitory control, attention shifting). Additional aspects of child development that merit consideration are abstract thought, the ability to take the viewpoints of others into account, and fine motor skills. Our goal is to align the design of the experiment with these competencies.¹⁸ If a child is unable to process a concept or do a calculation required, this will generate noise in the data and make it difficult to assess the underlying preference; in fact, in such situations no optimization principles may underlie even the most straightforward of choices, rendering an experimental exercise difficult, if not impossible.¹⁹

Here, we outline five competencies that we consider relevant for conducting economics experiments with children. We focus on the development of competencies from ages 3–18.²⁰

¹⁸While we focus attention on traditional experimental approaches with children, it is instructive to consider the work of Ye et al. (2021), which highlights the demands when using an electroencephalogram (EEG). In this case, the situation demands that the child sits still for a certain amount of time with the EEG nodes on, is not easily distracted by noise such as traffic, fully understands the task, etc. If using children that are too young, the attrition in such work will be substantial, therefore limiting the usefulness of data collected since the discard rate is so high. Broadly, this study highlights, in an extreme case, the importance of aligning child competencies to the demands of the experimental task.

¹⁹An exception to this is if the goal of the study is to examine the ability to perform a complex task.

²⁰Other social scientists have developed methods to evaluate children at even younger ages. For example, eye tracking and measures of attention can be used with infants (e.g., Meristo and Surian 2013). However, beyond work with parents and how their inputs into infants map to

1. *Numeracy*: Numeracy is the ability to understand and work with numbers. Foundational numeracy allows children as young as age three to compare groups of items and understand whether groups contain equal or different amounts of numbers. Formal set knowledge—that is, counting each item only once, counting in order, and stopping at the end of the count—develops later in life (Gelman and Gallistel 1978). Before young children have set knowledge, they may use shortcuts to divide items, for example by using a turn-taking heuristic to split a pile of prizes equally between two people.

The takeaway here is that experiments with young children should be designed to appeal to foundational, and not set, knowledge. For example, by age three, children should be able to identify that larger rewards are better than smaller amounts. This is of great import because maintaining control within the experiment requires that the reward medium is not only salient, but that utility is increasing in the reward (Smith 1982). At the same time, they may not be able to accurately count how many rewards they have received. Hence, in a dictator game, rather than asking children to report how many of a set they want to share with another person, experimenters should give children a plate of items and tell them to divide them as they wish, without using numbers. This general approach helps to achieve dominance within the experiment, a condition whereby the reward structure dominates the subjective cost of the activities of the experiment (Wilde 1980). Further, the literature teaches us that using familiar objects that are similar and arranging objects in lines with similar spacing can help children with making

comparisons (Carper 1942, Potter and Levy 1968; Tan and Bryant 2000; Clements, Sarama, and Macdonald 2019).

Importantly, one must distinguish the ability to count from set knowledge—just because children can count, it does not mean that they have set knowledge that allows them to count items in an experiment accurately. Set knowledge can be tested by asking children to share “X items” and observing whether the correct number is shared. As children get older, they evolve from being able to count two–three items reliably to being able to count more. Set knowledge (up to 10–20 items) and basic addition and subtraction are part of the core mathematics standards for five-year-old kindergarteners in the United States, and therefore by age six, children should become more savvy in their ability to manipulate up to 20 items in an economics experiment (California Department of Education 2013). By first grade—around age seven—students should be able to extend this knowledge to 100 items. Note that addition and subtraction lag counting—children can typically count to higher numbers earlier than they can add or subtract (Zur and Gelman 2004; Campbell 2005).

The next numeracy milestone relevant to economics experiments is understanding multiplication, division, and fractions. This more formal knowledge is useful for, for instance, explaining public goods games that vary the marginal per capita return (MPCR) that determines payoffs. The formal understanding of multiplication and division is expected by third grade—around age nine. But, the next several years are spent on solidifying this knowledge and learning to perform more complex manipulations of fractions. However, even if children do not have a formal understanding of division yet, they are able to understand how a set of physical items can be distributed among themselves and others. Hence, using physical props and familiar language can overcome this hurdle.

certain outcomes (such as, e.g., List, Samek, and Suskind 2018; Leung, Hernandez, and Suskind 2020), we did not find any economic experiments using infants, therefore, we limit our discussion to ages three and up.

The final useful numeracy concept is understanding of probability. In the United States, the Common Core math standards expect this concept to be formally introduced in sixth grade—around age 12—with more in-depth statistics instruction following in grades seven and eight—ages 13–14 (California Department of Education 2013). However, children at younger ages than 12 are exposed to—and likely understand—the basic concept of probability. Efraim Fischbein is credited with early work on the development of probabilistic thinking. His novel experiments evaluated the ability of children as young as age three to reason about probabilities and fractions (Fischbein 1975). His work suggests that intuitive reasoning about probability and chance develops early, and he finds that in some cases children ages 3–5 perform better than children ages 12–14 when evaluating the chances of events (Fischbein 1975). Further, children older than 14, as well as adults, may have difficulties with probability concepts. For example, research documents persistent overestimation of small probabilities and underestimation of large probabilities among adults (Kahneman and Tversky 1979).

A potential path to harness intuitive understanding of probability in economics experiments is to use spinners—wheels with differently colored slices—to explain concepts of risky options. This is done in the early work of Harbaugh, Krause, and Vesterlund (2002) and in many other experiments that we reviewed. Physical examples that are used in laboratory experiments with university students can be applied to experiments with children. For example, Halevy (2007) uses bins with balls to explain ambiguity and risk in laboratory experiments. The use of physical bingo cages is also popular in laboratory experiments and can be used with children (Harbaugh, Krause, and Vesterlund 2002; Harrison, Lau, and Rutström 2007; Tanaka, Camerer, and Nguyen 2010).

2. Literacy: Literacy is the ability to communicate using language. For researchers working with young children, an important distinction is productive vocabulary (spoken language) versus receptive vocabulary (understanding when spoken to). Receptive vocabulary develops sooner than productive vocabulary, meaning that at any given time, children can understand more than they can speak. Young children who cannot speak many words may still be able to respond to the experimenter through pointing and other gestures. This represents a key impetus as to why we suggest one-on-one implementation and providing physical props for children below age five.

Another experimental milestone in children is reading and writing, since this allows researchers to give subjects the independence to read instructions and mark their preferences. Most children learn to read by age seven, with fluent and independent reading emerging around age nine (Morphett and Washburne 1931, Fitzgerald and Shanahan 2000). Unlike with numeracy, there is no intuitive reading or writing ability that emerges earlier than this, though of course some children master reading and writing at earlier ages. We therefore do not advise conducting experiments that require substantial reading or writing with children below the age of 10.

The good news is that there are many alternative options. One alternative is instructions that involve pictorial representations (e.g., pictures of smaller amounts of candies today and larger amounts of candies tomorrow, in a time preference task). In such instances, children can circle which of the options they prefer. However, very young children (below age five) may still struggle with this task since their fine motor skills may not be sufficiently advanced. Hence, another alternative is electronic instructions (delivered on a computer or iPad) with pictorial representations that

allow for clicking or tapping responses on a screen.²¹

In sum, to maintain experimental control we suggest keeping language simple no matter the age of the respondents. Research shows that adults in America read at an eighth grade reading level on average (Davis and Wolf 2004). Health care is one area where readability of materials is particularly important; hence, health-care organizations provide guidelines for patient materials. A variety of national organizations have suggested that patient materials for adults be at or lower than a fifth, sixth, or eighth grade reading level (see Badarudeen and Sabharwal 2010 for more discussion).

3. Executive functions: Executive functions are higher-level cognitive abilities that encompass cognitive flexibility, attentional control, inhibitory control, and working memory (Goldstein and Naglieri 2014). These abilities cut across many aspects of experimental design. Cognitive flexibility is the ability to switch between tasks or switch between different aspects of a problem. Attentional control is the ability to avoid distractions. Research suggests that these skills develop with age (Dajani and Uddin 2015). We find it difficult to provide guidelines for how to design studies that incorporate this developing ability into experimental procedures. However, a useful rule of thumb is to avoid requiring children to “switch” their attention throughout the study: this is why, for example, when we conduct experiments at CHECC we are careful to use differently colored props for each decision task to avoid requiring children to work to switch their definition of a particular set of props (e.g.,

we use red/blue plates in a dictator game, and then use green/orange plates in a time preference elicitation task if it is in the same session.)

Working memory is the ability to consider and manipulate multiple facts simultaneously. Research on memory suggests that auditory memory is separate from visual and tactile memory, and that recall is better when information is presented via visuals (Gathercole et al. 2004; Cohen, Horowitz, and Wolfe 2009; Bigelow and Poremba 2014). This suggests that experimental instructions should be presented visually when possible. Research shows that the main components of working memory are in place by age six, and that their capacity increases linearly through early adolescence (Gathercole et al. 2004). Research requiring children to memorize numbers finds large increases in working memory with age, whereas research requiring children to memorize familiar objects documents a much more gradual development (for a summary, see Schneider 2011). This follows because familiarity with numbers develops concurrently—children asked to memorize numbers perform poorly on the task if they are unfamiliar with them. The takeaway here is that familiar objects and representations should be used when possible in economic experiments.

A related concept is focus and attention span. How long should experimental sessions reasonably last, based on the child’s age? We were unable to find any direct guidance for this, partly because how attention is measured affects the results (in the parlance of hard scientists, the “Heisenberg uncertainty principle” is at work, or in psychology the “observer effect” if the act of observation itself matters). Educators have developed rules of thumb based on some function of the child’s age; for example, multiply a child’s age by two, and this gives you the number of minutes that a child can pay attention. This implies that a five-year-old

²¹ To elicit risk and time preferences of their elementary school sample, Brocas et al. (2019) used electronic tablets programmed with Matlab. In this study, children viewed vignettes and made their choices by touching their preferred options on the screen.

can usefully participate in a study for 10 minutes, a 10-year-old for 20 minutes, and so on.²² Until further research is conducted, we cautiously endorse this rule to maintain experimental control, though we suspect the rule should be nonlinear, especially during the adolescent years. Brocas and Carrillo (2020) suggest a maximum limit of 40–50 minutes for an experiment for children up to middle-school age.

Of course, the situation itself will matter as well, making it equally important to consider how to sustain attention. The literature reveals that children can pay attention longer if there are fewer distractions (Dixon, Salley, and Clements 2006; Fisher, Godwin, and Seltman 2014; Gaspelin, Margett-Jordan, Ruthruff 2015). Hence, minimizing auditory and visual distractions is key for increasing the attention span of child participants. Another approach that works well is to take frequent breaks to engage in some physical activity, that is, asking children to stand up and do a few jumping jacks (Ólafsson, Livingstone, and Haddon 2013), and to keep children engaged by making sure that the instructions are interactive, that is, by asking the child's input while proceeding through instructions (Morgan et al. 2002, Punch 2002).

Inhibitory control is the ability to control impulses. The now well-known “marshmallow task”—where children who can avoid eating a marshmallow for some period of time receive two marshmallows—evaluates this skill (Mischel and Ebbesen 1970). More recent research in neuroscience suggests that inhibitory control is associated with future-oriented behavior in children (Steinbeis and Crone 2016). Our research suggests that inhibitory control can be decoupled from time preferences by incorporating a front-end delay for rewards in the experiment (Castillo et al.

2020). More generally, experimental designs that trigger an impulse to make an immediate choice should be avoided, since we generally want subjects to make deliberative decisions that are not driven by self-control issues. This is a key assumption to maintain dominance in the experiment.

4. Abstract thought: Abstract thought is the ability to think beyond the concrete, for example, thinking about possibilities and forming new ideas. Abstract thought develops throughout the 3–17-year-old period and affects children's understanding of anonymous partners in a game or games involving possible outcomes that are difficult to represent with visual aids. Children begin to reason using abstract and symbolic thought as early as ages three–five, when they use ordinary objects in pretend play (Bergen 2002). This suggests that children could understand anonymous partners at an early age. However, the abstract thought needed to represent concepts like probabilities is more challenging. For example, research shows that over- or underestimation of probabilities can be due to concrete, rather than abstract, thinking (Lerner et al. 2016).

The ability to think about future events is part of abstract thought. Research suggests that three-year-olds have trouble describing planned future activities, but that five-year-olds are much better at this task (Busby and Suddendorf 2005). In experiments eliciting time preferences, researchers have choices about how far in the future payments should extend. Some evidence suggests that children ages three–five perceive a one-day delay as “long” (Schwarz, Schrager, and Lyons 1983). Hence, Andreoni et al. (2019) use a one day delay for children ages 3–12. Most research with older children, however, uses longer delay periods on the order of four–eight weeks (Bettinger and Slonim 2007; Angerer, Lergetporer et al. 2015; Castillo et al. 2011).

²²See, for example, <http://www.laurenghill.com/understanding-the-of-attention-spans-of-elementary-aged-students/>.

5. *Theory of mind (ToM)*: The ability to separate one's own information set and desires from that of others—or ToM—is crucial for most social interactions and is relevant for many experiments we may consider conducting with children. Basic ToM usually develops around age four. Prior to this age, children have difficulty understanding that someone else's viewpoint can differ from their own, which makes games with different information sets unreliable for this age group. By age five, normally developing children have developed ToM, though higher-order thinking skills continue to develop through young adulthood (Wellman, Cross, and Watson 2001). Within our CHECC data we find that, for three–six year olds, making the presence of other children salient increases their ToM (Charness et al. 2019). This is important experimentally since there is some evidence that children from lower SES families lag children from higher SES families (Devine and Hughes 2018).

Evaluation of Competencies.—There are at least two reasons to evaluate child competencies. First, as discussed in subsection 10.4, researchers may be concerned that concurrent development in cognitive abilities may create confounds and may therefore want to control for cognitive abilities. Standardized assessment tools are available for this purpose. Proprietary tests, such as the Woodcock–Johnson Test of Achievement (WJ), include subtests for numeracy (e.g., applied problems, quantitative concepts) and literacy (letter–word identification, spelling) that can be used with children ages three and up (Woodstock, Mather, McGrew 2001). The Peabody Picture Vocabulary Test (PPVT) is a good measure of receptive vocabulary (Dunn and Dunn 1965, Williams 1999).

A highly recommended test for executive functions and other cognitive skills is the National Institutes of Health (NIH) Toolbox,

which is administered on an iPad (Gershon et al. 2013; www.nihtoolbox.com). The NIH Toolbox includes subtests for cognitive flexibility, attentional control, working memory and language. Additional constructs include processing speed, immediate recall, and episodic memory. Most tests described above require one-on-one administration with a trained interviewer. If researchers have access to grades or standardized test scores, these can sometimes be used as proxies for cognition, since these are often correlated.²³

A second reason for measuring child competencies is that researchers may be interested in learning whether children can solve a specific problem or understand a specific concept in their experiment. That is, they may wish to know whether behavior in their study is due to an inability to understand a task or to true underlying preferences. Likewise, such an exploration can yield insights on the cognitive effort children must make to complete the exercise. This understanding can help to ensure experimental control by providing an indication of the level of rewards one should use to ensure salience and dominance (Smith 1982). This would not require a comprehensive assessment. Instead, researchers should develop their own questions. We discuss the use of such comprehension checks in subsection 9.2.

9.2 *Experimental Design by Child Development Age*

Table 1 provides a comprehensive summary that describes the range of methods that have been used in experiments with children and adolescents. The papers summarized in this table come from an extensive search of the literature using economic experiments with children, including papers from other

²³Access to these materials has historically required providing evidence that the researcher has reasonable qualifications.

TABLE 1.
ASPECTS OF EXPERIMENT DESIGN BY CHILD DEVELOPMENT AGE

Developmental Stage (Ages)							
Experimental Aspect	Preschool (3–5)	Early Elementary (6–8)	Late Elementary (9–11)	Early Adolescence (12–14)	Late Adolescence (15–17)		Standard Practices
<i>Instruction Delivery Format</i>							
One on one	55.96%	43.90%	39.31%	32.48%	35.48%	Group	
Group	42.20%	53.66%	59.31%	63.25%	61.29%		
NP	1.83%	2.44%	1.38%	4.27%	3.23%		
N	109	164	145	117	93		
<i>Identity of Partner/Group</i>							
In person	33.04%	30.00%	26.79%	25.23%	23.08%	Anonymous	
Nonhuman	1.74%	3.00%	3.57%	2.70%	2.56%		
Photo	15.65%	11.50%	7.14%	0.90%	0.00%		
Gender	13.04%	9.00%	6.55%	5.41%	3.85%		
Age	13.04%	16.50%	15.48%	17.12%	16.67%		
Anonymous	26.09%	30.50%	36.90%	47.75%	47.44%		
Other	1.74%	9.50%	8.33%	7.21%	12.82%		
N	115	200	168	111	78		
<i>Instructions</i>							
Electronic	1.83%	3.66%	3.45%	5.98%	13.98%	Experimenter reads instructions out loud	
Physical	8.26%	4.27%	2.07%	2.56%	2.15%		
Written	0.00%	0.61%	2.07%	5.98%	7.53%		
Oral	88.99%	90.24%	91.03%	82.91%	75.27%		
NP	0.92%	1.22%	1.38%	2.56%	1.08%		
N	109	164	145	117	93		
<i>Instruction Aids</i>							
Physical	31.65%	31.03%	24.89%	12.35%	8.11%	None	
Graphical	23.42%	24.90%	24.02%	34.71%	35.81%		
Videos	0.63%	0.38%	0.87%	0.00%	0.00%		
None	44.30%	43.68%	50.22%	52.94%	56.08%		
N	158	261	229	170	148		
<i>Subject Decision Format</i>							
Electronic	6.96%	9.96%	13.54%	22.35%	25.50%	Electronic (e.g., zTree), rarely reply on paper	
Physical	63.29%	46.36%	35.37%	22.94%	12.08%		
Written	3.80%	20.31%	29.69%	38.82%	48.32%		
Oral	24.68%	22.22%	20.09%	14.71%	14.09%		
NP	1.27%	1.15%	1.31%	1.18%	0.00%		
N	158	261	229	170	149		
<i>Comprehension Check</i>							
Electronic	2.63%	1.95%	3.13%	2.99%	4.03%	Typically yes	
Physical	10.53%	8.20%	4.02%	3.59%	1.34%		
Written	1.97%	3.13%	4.02%	7.78%	7.38%		
Oral	42.11%	38.67%	36.16%	26.95%	19.46%		
None	42.76%	48.05%	52.68%	58.68%	67.79%		
N	158	261	229	170	149		

(Continued)

TABLE 1.
ASPECTS OF EXPERIMENT DESIGN BY CHILD DEVELOPMENT AGE (*Continued*)

Experimental Aspect \ Developmental Stage (Ages)	Preschool (3–5)	Early Elementary (6–8)	Late Elementary (9–11)	Early Adolescence (12–14)	Late Adolescence (15–17)	Standard Practices
<i>Total Number of Subject Decisions</i>						
Min	1	1	1	1	1	
Max	50	100	54	320	320	~20-40
Mean	5.679	5.859	6.282	12.278	13.543	
SD	8.374	9.840	8.357	33.380	35.129	
N	158	260	228	170	144	
<i>Experiment Length</i>						
Min	10	0.25	3	5	5	
Max	120	120	120	120	120	~1-2 hrs
Mean	31.880	31.781	38.617	46.990	45.045	
SD	27.096	26.953	29.582	32.386	32.958	
N	158	261	229	170	149	
<i>Payout for Repeated Decisions</i>						
All Rounds	90.70%	73.33%	68.67%	57.01%	43.62%	One randomly selected, or All
One Round	9.30%	26.67%	30.00%	39.25%	55.32%	
Multiple but not all	0.00%	0.00%	1.33%	3.74%	1.06%	
N	86	150	150	107	94	
<i>Incentives</i>						
Token/Store	25.48%	36.92%	39.04%	28.40%	17.57%	Money (Cash)
Money	1.91%	8.85%	19.74%	47.93%	68.92%	
Gift Card	0.00%	0.00%	0.88%	5.92%	6.76%	
Candy	24.20%	21.54%	17.11%	7.10%	2.70%	
Food	7.64%	4.62%	4.82%	1.78%	3.38%	
Toys	9.55%	8.46%	8.77%	3.55%	0.00%	
Stickers	29.30%	14.62%	7.89%	1.78%	0.00%	
Stationery	6.37%	4.62%	2.63%	2.37%	0.68%	
Other	0.00%	0.38%	0.00%	1.18%	0.00%	
N	158	261	229	170	149	
<i>Exchange Rate</i>						
1:1	51.02%	30.83%	18.49%	23.17%	18.33%	Varies
1 to many	20.83%	46.67%	59.66%	60.98%	63.33%	
NP	23.40%	20.83%	21.85%	15.85%	18.33%	
N	49	120	119	82	60	
<i>Game Type</i>						
Risk	9.49%	10.34%	11.35%	21.18%	29.53%	All
Time	12.03%	9.58%	10.92%	9.41%	13.42%	
Dictator	39.87%	35.63%	36.24%	25.88%	19.46%	
Strategy	16.46%	16.86%	14.85%	11.76%	8.05%	
Competition	3.16%	4.60%	4.37%	9.41%	10.07%	
Ultimatum	5.06%	6.13%	5.24%	6.47%	5.37%	
Other	13.92%	16.86%	17.03%	15.88%	14.09%	
N	158	261	229	170	149	

Notes: N = number of children, NP = not provided, and SD = standard deviation.

fields that use incentivized economic games. The appendix describes our search process in greater detail. We identified 257 papers for inclusion in the table. The list of papers can be found in List, Petrie, and Samek (2022). Nearly half of the studies were conducted in Europe, about a third were conducted in North America, and most of the rest were conducted in Asia and Africa.

We split the table's contents into five age categories: ages 3–5 (roughly translating to preschool age in the United States), ages 6–8 (early elementary school), ages 9–11 (late elementary school), ages 12–14 (early adolescence) and ages 15–17 (late adolescence). We do not include any studies of children below age three because there are very few (we identified only eight) and because we believe that children below age three generally do not have the competencies needed to participate in an economics experiment as we define them here. We do not include statistics from any studies of participants ages 18–22, since experimental economists have been conducting studies with university students for the past several decades. Instead, the last column of table 1 provides the generally accepted standard for experiments with university students. Note that neuroscience research concludes that the brain does not fully mature until the mid-twenties (Pujol et al. 1993). How young adults differ from older adults is also an interesting research question, but not one we address in this paper.

Below, we describe how features of experimental design in our review of the literature broadly change as we move from younger to older ages. We also compare the methods used with children to methods typically used in laboratory experiments with adults. This provides a summary rather than a set of recommendations, though as we outline below, many times the commonly used method in each age category is also most appropriate given the child competencies discussed in subsection 9.1.

Instruction Delivery Format.—This refers to the number of children participating per experimenter (one on one or in a group). 56 percent of experiments with three–five-year-old children use one-on-one implementation, while this changes to small groups and to larger groups with age (e.g., 61 percent of 15–17 year-olds are in a group). By comparison, laboratory experiments with university students include a typical session size of 20–30 students, limited by the number of computers in a standard lab. Given the limited ability of children to focus attention and the ease of being distracted discussed in subsection 8.1, as aforementioned, we believe that one-on-one administration for younger children is appropriate. Yet, this approach does not come for free. Such one-on-one administration leads to the assessor potentially playing a much larger role in the experiment. With young children, instructions are spoken and potentially not read verbatim, responses are usually verbalized instead of written, and with the one-on-one temporal constraints, if there is variation in assessor quality, different subjects are receiving different versions of the treatment, violating the stable unit treatment value assumption (SUTVA). We return to this issue in section 10.

Identity of Partner.—The identity of the partner can be relevant, especially for young children who may not have sufficient skills to employ abstract thinking skills. Hence, for children ages three–five, it is common to see the partner in person (33 percent) and relatively less common for the partner to be anonymous (26 percent). Alternatively, for later ages it is most common to have an anonymous partner (about 50 percent of the time for ages 12–17) and indeed this is also the norm for standard laboratory experiments that use undergraduates as subjects.

Instructions and Instruction Aids.—This category refers to how instructions are

presented to participants. Instructions are read aloud most (75–91 percent) of the time for all ages, and this is also true for standard laboratory experiments, where instructions are usually read aloud to a group while participants follow along. Physical props—for instance, plates/envelopes as containers of items for recipients, or physical tokens to be allocated among recipients—accompany instructions less than a third of the time among ages 3–5 and 6–8 and about a quarter of the time for ages 9–11. Physical props are rarely used for adolescents, and rarely used in laboratory experiments with adults (with the exception of bingo cages to reveal payouts used in some lab games).

Graphical aids are used in a quarter of the cases for ages 3–11, and more frequently—about 35 percent of the time—for children ages 12 and up. Such aids are useful to harness foundational math knowledge discussed in subsection 9.1. Electronic instruction delivery is rarely used, but when used it is a more common format for older children who are more capable of using these devices independently. Overall, electronic devices are utilized roughly 5 percent of the time for children below age 14, and used in 14 percent of the experiments for children ages 15–17. Since electronic devices have become cheaper and easier to program over time, the rate of their usage has increased significantly since 2015. In experiments from 2015 onwards, electronic instructions were used in 80 percent of the experiments for ages 15–17.

Subject Decision Format.—Subjects can make decisions in an experiment by pointing or moving objects (physical), verbally (oral), in writing by circling a response or writing in a response (written), or by clicking a button on a computer (electronic). About two thirds of three–five-year-old experiments feature physical subject decision format, which is important here because as we discussed in

subsection 9.1, receptive vocabulary emerges before productive vocabulary. Moreover, a written format is rarely used until subjects are at least six years old. A written format is used in only about 25 percent of experiments with children below 12, and in only about 4 percent of experiments with children ages three–five. This is also reasonable given that writing abilities are only emerging at early elementary age. By comparison, for adolescent subjects ages 12–17, decision format is more likely to be written (about 40–50 percent) or electronic (about 25 percent). This mirrors closely the standards for laboratory experiments, which are either conducted using paper and pencil with written responses, or conducted electronically with ZTree (Fischbacher 2007) or similar software (e.g., Matlab, such as in Brocas et al. 2019).

Comprehension Check.—We identify whether there was a comprehension check before proceeding to the decision, which could be oral, physical, written, or electronic. Comprehension checks are often used in standard laboratory experiments and should be even more important for young children, given the uncertainty about what they are able to understand. Surprisingly very few studies across age groups mention comprehension checks—comprehension checks are documented in only 30–60 percent of experiments across all age groups, with seemingly no pattern by age. A potential reason for this result is that comprehension checks are used, but not mentioned in the description of the experiment; or that given the shorter session times, they are not used in order to save time. The comprehension check strikes us as a valuable addition to any experiment with children—especially given the limited cognitive abilities of children relative to adults—and we therefore suggest that experimenters use this tool and report the results of comprehension checks in their papers.

Total Number of Subject Decisions and Experiment Length.—This category refers both to the time subjects spend in the experiment (in minutes, with minimum, maximum, mean, and standard deviation) and the number of decisions that subjects make in the experiment (again, with minimum, maximum, mean and standard deviation). Among children ages 3–11, subjects make 9 decisions on average. This increases to 34 decisions for ages 12–17. Experiments take 32 minutes on average for children ages 3–8 and 40–50 minutes on average for children ages 9–17, all of which exceed our recommendations on experimental duration discussed above, but falling within the Brocas and Carrillo (2020) recommendation to keep experiments a maximum of a class period in length. By comparison, laboratory experiments with adults tend to last 45 minutes to 2 hours.

Payout for Repeated Decisions.—In cases where subjects make many decisions, it is common in standard laboratory experiments to pay out only one randomly selected decision to avoid confounds caused by paying out all decisions, such as income effects and portfolio effects. But children may have difficulty understanding the concept that only one decision is paid. Indeed, over 80 percent of experiments in the three–eight-year-old age group pay for play in all rounds. We find that it is more common to pay only one round at older ages, with nearly 50 percent of experiments paying for choices in only one round at ages 9–17. Paying for choices in one round, while a common strategy in typical laboratory experiments, is less common in experiments with children. This is reasonable, since as we noted in subsection 9.1, children have difficulty understanding probabilities and paying out one round adds an additional layer of complexity, in some cases likely causing dominance or salience to be violated.

Incentives.—We identified four main approaches for incentivizing children in the literature: giving out a preset prize such as candy, a sticker, or small toy; giving out tokens that can be exchanged for prizes in a “store”; using gift cards; and using money. Our survey of the literature finds that common prizes include packets of Skittles, crayons, and balloons for ages 3–11 and chocolate bars, pens, and markers for ages 12–17. The appendix provides a detailed list of the prizes used by age group (see table A1).

As shown in table 1, for children ages three–eight, a prize store is the most common incentive (about 35 percent), and candy is the second most common (about 20 percent). The store is also the most popular at ages 9–11 and is used nearly in half of the studies. In some experiments that use a prize store, the prizes are revealed to the children before they play the game (Sutter et al. 2018, Brocas et al. 2019), and in others they are simply told they will get rewards but are not told what they will be (Bügelmayer and Spieß 2014). The former approach seems more appropriate to insure that the incentives are salient. The second most popular approach at this age is to use money (about 20 percent). After age 11, money becomes more popular and is used in 47 percent of the cases for ages 12–14, and 69 percent of the studies for ages 15–17. Gift cards are less common and are used in only 6 percent of the studies for ages 12–17. By comparison, the standard in laboratory experiments is to use money. In our view, this difference in payoffs over the age range is reasonable, since the payment medium should yield a salient reward where the subject’s utility is increasing in the reward and non-satiation holds. Of course, incentives are a cornerstone of economics and therefore deserve more careful attention. In subsection 10.7, we highlight important considerations when using incentives with children.

Exchange Rate.—When an exchange rate from points earned in the experiment to prizes or money is used, it is most common for this exchange rate to be 1:1 at ages three–five (51 percent) and to be more complex for ages six and up (50–70 percent of the time, increasing with age range). Exchange rates in laboratory experiments also tend to be more complex than exchanges in experiments with children.

Types of Experimental Games.—The last row of table 1 provides information about the types of games used in the experiments that we surveyed. A large proportion of economics experiments used with young children ages three–five were dictator games (40 percent), and this is partly because this is one of the easiest games to explain and partly because of the rise in popularity of dictator games among developmental psychologists, who are interested in the evolution of social preferences at a young age. Starting at about age nine, many standard economics games become well-represented (e.g., games eliciting time, risk, and competitive preferences; games of strategy; ultimatum games).

Comparing Methods.—A limited set of papers investigate how design features affect children's decisions. Bunch, Andrews, and Halford (2007) test a variation of the children's gambling task, wherein they introduce either a "gains only" or a "losses only" deck against a variable deck. With this variation, they lower the relational complexity and find that all children succeeded in these tasks. Angerer, Lergetporer et al. (2015) compare two methods of time preference elicitation in a large group of children, finding that a multiple price list and a single choice measure yield similar results in the aggregate. We urge future research to explore such methodological considerations, as a maturation of this field will depend on such new insights.

With the extant literature and its many heterogeneities in mind, we turn to a set of 10 tips that serve as guidance for the researcher interested in conducting field experiments with children. The tips might also help consumers of this literature more deeply understand the various design choices and their consequences.

10. *Ten Tips for Pulling Off Experiments with Children*

Sprinkled throughout the previous sections are numerous experiments conducted with children as subjects. The experiments sometimes leverage randomization for identification of the key treatment insight, sometimes leverage natural variation by judiciously applying experimental control in that setting to obtain preferences or other information to inform a model, and sometimes do both. Regardless of approach, the literature teaches us that similar to the spirit in which astronomy draws on the insights from particle physics and classical mechanics to make sharper inference, experiments with children can provide principles to sharpen inference around economic issues of import. Likewise, such experiments can help to determine whether received insights should be reinterpreted or defined more narrowly than first believed.

Our own experiences with experiments with children range from baseball card shows in the early 1990s to open air markets in the early 2000s, from starting our own schools in Chicago and Bangladesh to lab experiments in several parts of the United States. We have learned that the notion of conducting controlled studies with children might appear daunting. Figuring out the market, how to randomize relevant features within a complex situation, and avoiding pitfalls is not taught in standard economic or even social science courses.

In this section we offer 10 tips for executing successful experiments with children. We suggest that all economists conducting experiments with children carefully consider these tips, which range from tips about idea generation to avoiding pitfalls that are present in much of the current literature, and practical considerations such as recruitment and incentives. These tips were borne out of our interpretation of where the literature with children should be taking us, as well as discussions with colleagues who had practical questions about how to conduct experiments with children. These tips are not intended to be treated as universal truths; rather, they are intended to help researchers think through the design and implementation of experiments with children.

The first two tips drive home the point that it is important to consider theory and mechanisms when developing research agendas using children. The next two tips remind researchers to be cognizant that children are not just “little adults.” Children develop rapidly in childhood and adolescence. It is critical to incorporate measurement of this development and to acknowledge potential limitations in interpreting findings due to this development. The fifth tip straddles interpretation issues and practical design issues. It focuses on recruitment, and in particular, reminds researchers to seek representative samples or to be cautious of drawing broad conclusions from nonrepresentative samples. The next four tips provide practical advice related to procedures, incentives, ethics concerns and documentation. Finally, the last tip discusses scaling concerns, which is relevant for researchers designing interventions for children.

10.1 *Use Economic Theory to Guide Your Design and as a Lens to Interpret Your Data*

This tip is ubiquitous, and any researcher interested in experimentation should begin

here. Indeed, this is tip #1 from List's (2011) 14 tips for pulling off a controlled field experiment. Yet, it bears repeating here since economic theory is portable. By themselves, data from experiments with children can only offer a limited understanding about what is likely to happen in a new setting or with a new population of children. Incorporating guiding principles or hypotheses that drive the research agenda—whether this is a formal theory or an informal conceptual framework—is key for guiding the interpretation of results. Together, theory and experimental results provide a powerful tandem to learn about the world and provide prescriptions for improvement.

10.2 *Go Beyond A/B Testing to Deepen Our Understanding of Children*

A natural starting point from 10.1 is to explore mediators to observed relationships. Such mechanism exploration begins with theory, which is a key guide to determining a path deeper than simple A/B testing. There is a large literature in human capital development that relies on A/B testing—that is, comparing educational interventions to control groups—to draw inferences relevant for public policy (e.g., Heckman et al. 2010; Attanasio, Meghir, and Santiago 2012; Sylvia et al. 2021). This is an important approach that highlights the value of experiments with children. Throughout this article, we have also highlighted how we can use experiments with children to deepen our understanding of the underlying mechanisms behind the data patterns observed. In this manner, in addition to measuring the intervention effects (e.g., does this new kindergarten curriculum lead to higher test scores?), researchers should design sub-treatments tightly linked to economic theory to reap the true benefits of the experimental method by exploring why it works (e.g., a sub-treatment, or information gathered in a complementary survey, which

shows the channels through which the kindergarten curriculum worked). Getting to the whys of an intervention also have important effects on scaling, as discussed in subsection 10.10. Czibo, Jimenez-Gomez, and List (2019) provide a deeper discussion of going beyond A/B testing in broader economic settings.

In a nutshell, the combination of 10.1 and 10.2 provides a natural umbrella for a research agenda with children. As an example, if the researcher is interested in human capital formation, one first examines the optimal timing and type of investment over the life cycle, such as characterizing the types of interventions that are optimal. At that point, explorations going beyond A/B tests determine the important moderators and mediators of the examined interventions (see, e.g., the work at CHECC cited above using traditional experimental approaches and the work of Ye et al. 2021 for an example using EEG). With that knowledge in hand—the whys, wheres, whens, and for whom—applicable theories and optimal policies are better understood, and the basics for scaling are in place.

10.3 *Align Child Competencies to Experimental Design*

We next turn to issues that are specifically relevant to studies of children. The importance of aligning competencies to experimental design cannot be underemphasized, since children are not simply “little adults” and the developmental process is complex. This is why section 9 is devoted to describing how child competencies develop with age and the key design features that should be taken into account when designing experiments with children and adolescents. Aligning child competencies to experimental design requires an understanding of cognitive skills and executive functions. Table 1 in section 9, which summarizes how related

work has tackled this issue, can help guide the design of the experiment.

When studying how child preferences develop with age, competencies that develop around the same age range can be a confound. This is especially true for cognitive abilities. For example, research in developmental psychology shows that a young child's ability to split rewards equally is associated with math ability (Chernyak et al. 2016). Research in economics also shows correlations of cognitive abilities with child preferences (Benjamin, Brown, and Shapiro 2013; Andreoni et al. 2020), and suggests there is an association between ToM development and sharing behavior (Cowell et al. 2015). Competency development can also be related to SES since children from low-income households tend to have lower literacy and numeracy skills than their higher-income peers at the same ages (Liaw and Brooks-Gunn 1994; Noble, McCandliss, and Farah 2007). Moreover, consistent with our framework outlined in section 4, within the same age range and background, children can vary greatly in their level of cognitive development (Sternberg 2011). Hence, it is important to consider the associations between competencies and decision-making in the experiment. In the next tip, we discuss assessment of competencies as an important way to avoid confounds. Importantly, experiments that are not designed to be appropriate for the range of cognitive abilities in the study sample may introduce serious endogeneity issues if subjects who do not understand the experiment cannot participate due to lack of understanding.

As aforementioned, when we align child competencies with optimal design, we might be presented with unique problems. One such example is if one-on-one administration is necessary (as we suggest in section 9 for three–five-year-olds), and there is variation in assessor quality. In this case, a key assumption underlying the internal validity of the

experiment is violated: the SUTVA. This is because different subjects are receiving different versions of the treatment, rendering any simple comparison between treatment and control groups uninterpretable. We therefore urge the researcher to devote special attention to how assessors are allocated to subjects, making sure that assessors are distributed evenly across treatments, and examining if the assessor fixed effects are important to outcomes.

10.4 *Consider Stability and Confounds with Cognitive Development*

The timing of preference development is an important question. For example, when applying theories of rational decision-making to children, it is necessary to consider at what ages children can choose rationally. When designing interventions for children, one needs to have some idea of utility functions that would allow policy makers to assure that interventions are appropriate (e.g., providing immediate incentives for younger children, who are more impatient than older children). Finally, learning about how preference development is assisted by or interacts with cognition can give us a deeper understanding of what key milestones are needed for economic preferences to develop.

When evaluating the development of preferences with age, it is important to consider stability of measurement. For example, if economic preferences are unstable with age, does this mean that the underlying processes affecting this development are not stable or that the elicitation methods used are not stable? One way to evaluate stability while reducing measurement error is to elicit preferences using several methods (as discussed in, e.g., Gillen, Snowberg, and Yariv 2019; Angerer, Lergetporer et al. 2015 used two methods to elicit preferences, as do Datar, Nicosia, and Samek 2022; Samek et al. 2021) and account for decision error (e.g. Castillo, Jordan, and Petrie 2018). When eliciting

preferences over a large age range, different methods were often used with different age groups (e.g., as in Andreoni et al. 2019). When using different methods for different ages, it is useful to have overlap in the methods and age ranges to assure that there is a correlation with methods.

Cognition itself can also be an issue since it changes over time. Hence, if economic preferences are correlated with cognitive ability, but cognitive ability is not measured, then it is difficult to determine whether cognition is one channel through which age-related development affects preferences. Further, if researchers report that time preferences are correlated with a particular field outcome, it is unclear whether time preferences themselves or a correlated factor drive that relationship. Measuring and controlling for cognitive ability is one way to improve on existing methods; this was done in a number of CHECC papers (e.g., Cappelen et al. 2020, Castillo et al 2020). For example, Castillo, Jordan, and Petrie (2019) report that time preferences measured early in life are associated with later field outcomes, even while controlling for SES, cognitive abilities, noncognitive abilities and other economic preferences.

Measuring cognitive ability also helps with another problem, which is that lack of sufficient cognitive skills to understand a decision task can result in greater errors in the task at younger ages than at older ages, reducing the precision of the estimates at younger ages. One way to assess whether difficulties in understanding the task affect the results is to include questions on understanding in all experimental instructions (and to record the child's answers to these questions). Relatedly, insufficient noncognitive abilities could affect responses to a task. For example, Brocas and Carrillo (2018) found that children may have the ability to think strategically, but may not have sufficient noncognitive skills to act strategically.

Children who do not understand the experiment may either not respond or may provide random answers. Researchers may need to impute data or drop data in this case (dropping data seems to be most common). Such treatment of the data has a major limitation, which is that the subjects that remain would be a selected (higher cognitive ability) sample, which will introduce endogeneity and affect interpretation. A far better alternative is to create an experimental design that is suitable and easy to understand for the lowest-performing children in the sample. This can be achieved through following our guidelines in the prior tip and doing iterative pilot testing.

An area that has received less attention in economics than it has in psychology, psychometrics, or medicine is test–retest reliability (DeVon et al. 2007). Test–retest reliability is relevant for constructs that are not expected to change over time. Conducting experiments with the same group of children over a short time period would help to illuminate exactly how much error exists in the measures that we use (see Gillen, Snowberg, and Yariv 2019 for such an approach with adults). Designing batteries of tests all aimed at measuring a particular economic preference would help to reduce this error problem.

10.5 *Recruit a Representative Population*

This tip straddles issues of interpretation and practical concerns. With regard to interpretation, if the researcher wishes to make broad conclusions about children, it is important to either recruit a representative sample or to acknowledge the limitations in terms of drawing broad conclusions from a sample that is not representative. Most studies with children use selected samples. Unlike standard laboratory experiments with university students (also a selected sample), there are several ways in which subjects can be recruited, and in turn, several

ways in which child samples can be selected. Selection can occur on the basis of the SES of the participating school, buy-in from only a subset of parents, or even based on criteria for cognitive abilities required to complete the task. Depending on the research question, selection may be desired. For instance, a researcher wishing to study economic preferences of children with specific mental illness would wish to target recruitment to a sample of children that is representative of the population of children with mental illness. In what follows, we outline several popular approaches for recruiting children and discuss the benefits and drawbacks for each.

Recruitment of child populations is the first obstacle in conducting research with children. A popular approach in economics is to forge partnerships with schools and contact parents through these schools to invite participation (see, e.g., Angerer, Glätzle-Rützler et al. 2015; Angerer, Lergetporer et al. 2015; List and Samek 2015; Castillo et al. 2011; Houser et al. 2015; Harbaugh, Krause, and Vesterlund 2001). A major benefit of such an approach is that schools have large lists of eligible children, and parents may be more likely to participate if they receive the information from a trusted source. Further, schools can often benefit from the knowledge gained from the study. However, a potential drawback is that many schools are already overwhelmed with research requests and might not have the capacity to engage in another project. Also, some districts have additional application processes before allowing research with their students. If parent consent is necessary for the research, researchers should use a multipronged approach to engage parents, including sending flyers home in backpacks, as well as recruitment events on site and phone calls. This is because many parents will not return a consent form sent home in a backpack, but they may respond when provided with more information. If a school partnership is

possible, we believe this is one of the best ways to proceed due to the large potential sample size available and the collaboration between researcher and schools. Schools are also a good idea if researchers wish to follow-up with children at a later date, since schools could provide updated contact information for children if they have not moved out of the school district. Another advantage is the ability to link children with administrative data (e.g., grades, standardized test scores, or disciplinary referrals) with parent and student consent.

Some researchers cite difficulty in getting school buy-in as a major obstacle to recruitment and propose alternatives. An alternative approach that has been gaining popularity among developmental psychologists is conducting recruitment in public places, such as museums and public parks (see, for example, Blake and McAuliffe 2011). This includes setting up a table at a location and inviting families who pass by to participate. Museums sometimes charge researchers to use the space, while public parks are free (though may still require permission from the city). This method can be effective if the researchers can provide credibility, parents are not rushed, and their child can participate on the spot. But the drawback for researchers wishing to recruit subjects for an intervention or longitudinal study is that it could be more difficult to keep track of populations recruited this way. Another drawback is that if the research question involves a particular subpopulation of children (e.g., those with mental illness), then attempting to recruit in public spaces may be time consuming and suboptimal.

Another possibility is recruiting through existing parent networks such as Parent-Teacher Associations (PTAs) affiliated with nearby schools and asking them to forward information about the study to parents (Foss et al. (2011) cite this as one of their most successful approaches). This may

work better than recruiting at museums or public parks if researchers are seeking children with a particular characteristic (e.g., children of a specific age). Another alternative along these lines is to post messages in social media groups, which has the benefit of being able to select groups associated with a particular characteristic that is targeted (e.g., mental illness). This approach does require interested parents to follow through with bringing their child to the experiment.

Finally, one can obtain registry data to mail invitation letters to families. For example, county birth records are used to send mailed recruitment letters to parents of infants for some psychology studies (e.g., Kim and Johnson 2013). Registry data is also used in Deckers et al. (2015), who saw a recruitment rate of about 12 percent from their mailings (they offered participation in a free program as part of the study incentive). While we have been unable to find recruitment rates for related psychology studies, we suspect it to be substantially lower, as many of these studies do not offer incentives.

Having a representative sample of children in the study is important for drawing general conclusions or making claims of external validity. The registry data approach may give researchers the highest chance of obtaining a representative sample since it involves recruiting a wide sample of the population, though even then the participants who choose to participate are likely to be selected. Focusing the study on one school reduces representativeness since children within a single school are likely to be representative of only a single area, hence an alternative to improve representativeness is to select schools with children from a range of backgrounds.²⁴

²⁴An example of such an approach is in Brocas and Carrillo (2021), who conduct their study in a private school in a relatively rich area and a public school in a relatively disadvantaged area.

Suggestions for reaching out to parents include using simple language to communicate study goals (this can be very broad, e.g., understanding how children develop) and making study participation as convenient as possible (i.e., nearby study locations, catching the parent at a time they are not rushed and offering participation on the spot, providing flexible scheduling options). Offering an appropriate participation payment and gift for the child can also incentivize parents to participate.

10.6 *Adopt Appropriate Study and Experimental Procedures*

This tip and the following three tips deal primarily with practical concerns. An important consideration is where and how the research should be conducted. A standard approach in developmental psychology for research on children is to ask parents to bring the child to a child-friendly “laboratory” on campus. Such a facility should include age-appropriate seating, provisions for a parent waiting area, and adequate parking. Economists typically won’t have a facility like this. But the good news is that economic experiments do not require major equipment, so setting up a “mobile” laboratory is not difficult.

One possibility is to develop a partnership with a school, which is already child-friendly, and then either conducting experiments during class time or by asking parents to accompany children to the school in the afternoons or on weekends. Another alternative is to recruit children to participate at the recruitment location, for instance at public parks or children’s museums (more on this in the next tip). A major benefit of these alternatives relative to participation on a university campus is that it does not require the parents to travel very far to participate. However, a major drawback is that not all schools will be amenable to this approach—for example, poorly funded schools in low-income developing countries may be noisy, disorganized, and

violent. Another drawback is that even if the school is conducive to research, more extensive set-up time will be required at the field sites; for example, it is a good idea to try to remove visual distractions by putting up cardboard dividers and to look for areas that do not have highly distracting noise levels.

Another possibility is to conduct experiments in participants’ homes. In this approach, the experimenters (teams of two are a good idea for safety reasons) arrange and then conduct home visits. This approach is also not without drawbacks, since particularly in low-SES environments, homes may be overcrowded and full of distractions. Further, not all families will feel comfortable with researchers visiting their homes.

Since many of the experiments we describe (especially those with young children) rely on verbal and physical interactions, the capabilities of the experimenter are of vital importance. Experimenters should be experienced in interacting with young children. A number of training sessions specific to the protocol should be conducted to assure that the experimenter has memorized the instructions and can easily carry out the experiment. In our work, we often train undergraduate students at the university to conduct sessions. The university students we hire most often come from fields such as public policy, social work, and education.

Above, we have provided a number of alternatives for where to conduct experiments with children. We are not aware of data that explore the association of study environment with outcomes, but we think it is reasonable to assume that reducing the level of distraction in the environment will lead to higher precision of estimates.

10.7 *Understand How to Use Incentives to Motivate Children*

The topic of incentives arises most frequently when discussing design features of experiments with children. The pioneers

of the experimental approach argued that incentives are an important component of the experimental environment because of the control that they provide. In particular, the reward medium must satisfy three conditions: non-satiation, salience, and dominance (Smith 1982). As discussed earlier, cash incentives, which are typically used in laboratory experiments with university students, may not work for all age groups. Table 1 presents a progression of incentives by age, which shows that incentives at early ages most often take the form of candy or stickers, middle-childhood incentives often take the form of prizes, and adolescents can be compensated with gift cards or cash. This progression is appropriate, since children are not usually shopping on their own until middle to late adolescence.²⁵ When considering noncash rewards, many begin with attempting to solve heterogeneity issues. While heterogeneity over the valuation of the reward medium does not present problems of internal validity if the groups are balanced across treatment and control, the use of nonmonetary rewards does present unique challenges in other ways.

For example, consider the case of providing noncash rewards, such as stickers, to five-year-olds and asking them to play a dictator game. This is usually done as a means to measure social preferences among young children. The researcher wishes to interpret “number of stickers shared” as a social preference measure, yet such a measure is easily confounded with, for example, the child’s preference for the certain types of stickers in his/her endowment, by the child’s preference for stickers more generally, or by the child’s expectation about the preferences toward stickers of his or her partner. While

these issues are also present when considering the allocation of dollar bills of university students in standard dictator games (utility value of money is indeed heterogeneous), dollars can buy a myriad of goods and serve a useful intermediary role in modern economies. Alternatively, there is not a ready resale market for stickers, and therefore they are best considered for final consumption. As such, in experiments of this ilk, we learn a lot about preferences over certain stickers, but perhaps nothing more generally without further assumptions. Alternatively, when we use a medium of exchange such as currency as an incentive, we learn about preferences over that medium, and non-satiation, salience, and dominance are more likely to hold.

One way to begin to address this issue is by allowing the child to first choose stickers that he or she most prefers, as in Cox et al. (2016). Similarly, children may be asked to sort potential prizes in order of desirability, and the top few are chosen as the incentives for future games (Broome and Rice 2004). This approach assures that the children value the incentive. Another possibility is to use a prize store with prizes that are preselected based on desirability by the target group. Here, the children make decisions with tokens that are exchanged for goods in the prize store. Provided the prize store is stocked with goods that all children like, and they assume other children like them as well, then this is closer to using monetary payoffs, although the payoffs are made in lumpy, discrete amounts. There is a problem of potentially large diminishing marginal utility over two of a similar item; therefore the prize store should have a large enough variety to limit satiation.

The prize store is flexible, since researchers can include prizes, toys, school supplies, and so on that appeal to different age groups and tastes. The prize store is also easily explained—for very young children, a 1:1 exchange rate of points earned in the

²⁵Note that some local institutional review boards (IRBs) may not allow incentives, and that schools or other spaces where the experiment is conducted may have guidelines or limitations on types of incentives used.

experiment to prizes can be used, which limits the need for calculating prices of items. Using a prize store also allows for comparisons across age groups or for longitudinal data collection, since the store can be set up with items that appeal to all age groups. For example, in studies with children and their parents, children could earn points to exchange for small toys, while parents could earn points to exchange for small items like snacks, office supplies, or travel toiletries.

The other aspect that frustrates interpretation in experiments with children is that children are often rewarded by parents or teachers for doing what the adults want—and punished for doing what the adults do not want. Therefore, the potential for demand effects—especially when choices are not made privately—are potentially large. In addition, parents may expect children to share prizes with siblings after the experiment. Hence, measured preferences can be confounded with expectations about background consumption. The former issue can be minimized to some degree by allowing for private decision-making whenever possible. We have previously attempted to minimize the latter issue by including notes sent home to parents asking them not to require their children to share their incentive (although, such notes are more useful in repeated experiments, since in the first experiment children may expect a sharing requirement).

The problems raised in this section are more relevant for studies attempting to measure economic preferences than to studies considering treatment effects. Interpreting the level of an economic preference—for example, social preferences, risk preferences, or discount rates—relies on making assumptions about valuation of prizes. The issues are less problematic in experiments that seek to measure treatment effects if an assumption is made that valuation for incentives does not interact with treatments, provided salience, dominance, and non-satiation are satisfied.

In summary, we recommend prize stores be used for very young children whenever possible. Children begin having more independence over spending their own money to purchase snacks or toys by around age nine. If the researchers do not intend to compare to a younger age group, then money should be used from this point forward. Money could take the form of gift cards if the experiment is conducted in schools that prefer not to distribute cash or if the experiment is in the field and the experimenter(s) prefer to limit the amount of cash they are carrying at one time. We found that the most commonly used type of gift card varies by country. Walmart and Amazon gift cards were commonly used in the United States (Castillo et al. 2011; Castillo, Jordan, and Petrie 2018; Barash et al. 2019), while Bol.com vouchers were used in the Netherlands (Bekkers 2017).

10.8 *Obtain IRB Approval and Preregister*

This tip revolves around two key preexperimental administrative tasks that are invaluable. Indeed, this tip applies to experiments with children, but also spans all work that involves human subjects. We will discuss each in turn. In the United States and many other developed countries, the IRB or ethics committee provides oversight into research activities involving human subjects. Some US-based researchers may wish to conduct human subjects work in developing countries. They may therefore face additional regulations based on local laws and IRB requirements. In addition to obtaining relevant local permissions, researchers should still get IRB approval from their home institution. Much of the below provides guidance for the typical rules at US-based IRBs. Researchers based in or working outside of the United States should do their own investigation to determine what rules apply.

Children are considered a vulnerable population and require special additional protections. However, most IRBs are open to experiments with children, provided that the researchers have identified a clear rationale for why children are included in the study and have addressed ethics concerns specific to children and broader ethics concerns that would also apply to studies with adults. IRBs are familiar with research with children since many researchers in departments such as psychology and education conduct research with children.

Most studies described here will generally fall into the category of minimal risk. As such, the only additional consideration is that the consent process proceed in stages. First, the parent will be asked to complete a consent form on behalf of their child. Usually, the consent of one parent is sufficient.²⁶ Second, children will be asked to agree to participate (called “assent”) either verbally or on a written form. Each IRB may have their own requirements for assenting children, and they will usually consider the age and maturity of the child to determine (i) whether assent is required and (ii) whether assent needs to be documented in writing or can be elicited verbally.

Studies with children that provide an intervention or require longer-term follow-up have a more in-depth IRB process than one-off studies measuring an economic preference. For example, because the CHECC program featured an early childhood intervention administered by the researchers, as well as a consent form allowing for 14+ years of long-term follow-up, the CHECC study underwent full board review. This included a

presentation to the IRB regarding the nature of the study and the plans to minimize risks.

Just as with studies that use adult subjects, a key ethical consideration is that children do not feel coerced into participation—either from experimenters, teachers, parents or peers. One way to minimize potential coercion from teachers or peers is it have children choose whether to assent in private. To minimize potential for coercion from parents, stress to parents that it is up to the child whether or not he/she wants to assent and do not withdraw the parents’ participation payment if the child does not wish to participate.²⁷ And, incentive payments to children—which often take the form of gift cards or small gifts or candies—should not be so extreme that children feel compelled to participate or to share their payment with parents, siblings or friends.

In many cases, obtaining consent from the parents of all the children that are eligible for the study is challenging. For example, a researcher may be conducting a school-based intervention, but only some parents have consented for their child to participate. The most typical solution is to offer the program to all children in the study, but only evaluate or keep the data from those children who have consent on file. In other cases, for example if the research involves minimal risk and would be undertaken during the school day anyway, the IRB may allow the researchers to waive consent and instead provide parents with information sheets with the ability to opt out. In our experience, the proportion of consent forms that are returned is highly dependent on the SES of the school as well

²⁶As noted by a reviewer, some areas where research might be conducted may have low literacy rates or no written language. In these cases, written consent may not be practical, and the researcher should work with his/her IRB to develop appropriate consenting procedures.

²⁷There are also possible costs to paying parents based on their child’s performance in a task, and such interventions should only be considered after careful ethical review. For example, parents who do not receive an expected reward due to their child’s performance may retaliate against their child, which would be a concern to consider carefully.

as the buy-in from the teacher (who can nudge parents to return forms).

Subsection 10.5 discussed various ways to recruit participants. All of the proposed channels should be documented for the IRB to assure that they follow guidelines and are not considered coercive. For example, contacting parents or students through social media would require IRB approval. It is against the policy of most IRBs that we are aware of to provide referral bonuses or incentives that are linked to recruitment of subjects.

Finally, just as with adult subjects, a careful consideration of costs and benefits should be undertaken. Children should not be used as a convenience sample—that is, just because one can go into a school to conduct experiments with children does not mean that they should be used instead of adults. Child subjects should be used when the research question would benefit from the participation of children in the study, such as when the researcher is studying the development of economic preferences in childhood.

While IRB approval has been around for decades, a relatively new (at least for economics) preexperiment task of import is preregistration. Established in May 2013, the American Economic Association's registry for randomized controlled trials (AEA RCT Registry) provides a venue for researchers to document their experiments in a manner that is searchable by external audiences (Abrams, Libgober, and List 2020). Researchers document the specific details of the experimental design, target number of observations, et cetera, and can also include a pre-analysis plan detailing how the researcher plans to analyze the data.²⁸ In theory, if used appropriately, this innovation can tackle key issues in the credibility crisis, most specifically, the

“file drawer” problem and “p-hacking.” Both issues are key to solving the false positive problem. As Abrams, Libgober, and List (2020, p. 1) discuss, the registry thus far has had quite limited success: “over 90 percent of randomized control trials (RCTs) in economics do not register, only 50 percent of the RCTs that register do so before the intervention begins, and the majority of these preregistrations are not detailed enough to significantly aid inference.” While other registries show similar impotence, and therefore provide limited advice on improvement, as part of this tip we follow Abrams, Libgober, and List (2020) and urge experimentalists to at least submit their IRB materials in the preregistration *before* any collection of data.

10.9 Document Procedures and Promote Replication

Beyond attenuating false positives via preregistration, replication represents a key feature that can lower long-run false positives. Proper documentation of experiment procedures is key for future replication.²⁹ Laboratory experiments have benefitted from unwritten standards about what is important to report—and researchers who report on laboratory experiments typically provide information on session size, experiment length, average payout, how instructions were provided, whether privacy dividers were in place, and whether questions on understanding were used. Further, a standard in laboratory experiments is to provide experiment instructions in an appendix. We propose that similar standards be adopted for experiments with children, since many papers that we surveyed did not report on these key factors. For example, some papers failed to report on the session size or the exchange rate between tokens

²⁸A researcher can register a study without submitting a pre-analysis plan and can submit a pre-analysis plan without registering (although not on the AEA RCT site).

²⁹Further, proper documentation is also important for preregistration and pre-analysis plans, which are becoming necessary standards and conditions for publication.

received in the experiment to prizes in the prize store. Papers should also report on the environment in which the study was conducted—including details such as level of privacy and distraction—which could help researchers understand how important such details are for the measurement of outcomes. These variables are all important for future attempts at replication and for moving the methodological literature forward.

10.10 *Use Backward Induction to Design Interventions That Will Scale*

A complement to the representativeness of the population discussion in subsection 10.5 is the representativeness of the situation. Indeed, to take both discussions to a higher level is to consider whether your insights will scale. This tip is aimed at the first silo that deals with conducting experimental interventions. When doing that type of experimental intervention, researchers should always be aware of how our interventions scale or have effects at levels beyond the setting of the experiment. To be more perspicuous, consider the “voltage effect” described in List (2022). When considering the effects of programs, a common finding is that the treatment effect sizes diminish substantially when moving from small scale to a larger scale (see also Al-Ubaydli, List, and Suskinid 2017; Al-Ubaydli et al. 2017). We are only at the beginning of understanding the science of using science, but there is some evidence that shows there are good economic reasons why program effects fade at scale, and how benefit-cost profiles change at scale. Al-Ubaydli, List, and Suskind (2020) examine the market for knowledge and show that the incentives of the various actors (researchers, government policy makers, citizens) themselves can lead to the scale-up problem. Importantly, if researchers backward induct when setting up their original research plans to ensure accurate and swift transference of programs to scale, not only

are initial insights useful, but they also aid in the generality (external validity) problem. Al-Ubaydli et al. (2021) provide a checklist for both researchers and policy makers that involves design choice from idea inception to roll out at scale.³⁰

11. *Discussion and Conclusions*

Experiments with children are uniquely situated to provide insights to economists and policy makers. Our framework proposes that child outcomes depend both on genetics and environment, and that investments by parents, schools, and the children themselves affect these outcomes. The literature that we survey documents large heterogeneity in preferences even at an early age and finds evidence that parenting and school-based interventions have an impact both for the formation of child skills and economic preferences. Our model further suggests that child economic preferences affect both contemporaneous and future child outcomes. The literature that we survey estimates the economic preferences of children and links these preferences to SES, parents, peers, and field outcomes.

We view the main methodological takeaways from the experimental literature as follows. First, general patterns in economic preferences related to age or demographics have now been identified, especially for elicitation tasks that are relatively easy to explain (i.e., for dictator games, time preferences, and risk preferences). Second, links between some experimental tasks and some field behaviors have been established. This is good news since it means that methods currently used are mostly appropriate for

³⁰This is not the only consideration when conducting intervention studies. For example, interventions should be designed to be culturally appropriate, which can be achieved through iterative discussions with relevant stakeholders, e.g., families, teachers, and school administrators in the area where the intervention will be conducted.

measuring the intended constructs. An outlier is competitiveness—evidence in this area is more mixed and authors have attributed these results to differences across cultures or methods. Third, results on parental transmission of preferences and impact of environment are also mixed. A commonly accepted set of methods for conducting experiments with children will be helpful in reconciling some of these diverse results.

In the introduction to this manuscript, we discussed several questions that are important for silo 2—understanding children to understand adults. Namely, we asked about the association of preferences with family background, the role of experience in shaping preferences, and about when traits and preferences emerge and how they change over time. The literature to date has begun to provide answers to these questions. For instance, as evidenced by the research summarized in sections 5 and 6, we are learning that preferences are associated with family background but are also malleable in the sense that they evolve as children grow up. The research described in section 8 shows that inputs from external factors such as schooling affect preferences, in the sense that (some) preferences can be affected by (some) interventions. A remaining open question here is whether such interventions affect preferences themselves, or whether they affect the behavioral strategies that are used by the children. That is, if we use an intervention to affect willingness to wait for future rewards, have we changed underlying preferences or simply provided children with tools to use to help them wait for future rewards?

Further, we have begun to answer the question about whether parents and external influences affect preferences. It is clear that SES and parental preferences are critically important since research summarized in subsections 6.1 and 6.2 shows that both SES and parental preferences are correlated

with their child's preferences. In the case of parental preferences, the correlations are more commonly found in studies of older children and not strong in studies with younger children, suggesting that perhaps the transmission of preferences from parent to child takes time to manifest. This provides inferences into understanding why preferences of adults are widely heterogeneous. In particular, when considering inequalities by SES among adults, one important driver could be the differences in economic preferences manifested at an early age.

Finally, it is interesting to note that many preferences and behaviors are already present at a very early age. As discussed in section 3, very young children are able to make rational decisions and make trades in markets. Further, as discussed in subsection 5.4, the gender gap in competitiveness observed in adults emerges at an early age. Thus, there seems to be a component of economic preferences that arises early in life, even prior to formal schooling. This points to the importance of studying children prior to school age, which is also discussed in List, Samek, and Suskind (2018).

There are also a number of open questions in the related work. First, few papers focus on field outcomes that are long term. Instead, many of the existing papers discuss abstract topics or experiments in very controlled settings with contemporaneous or short-term outcomes. A deeper understanding of how preferences affect field outcomes in the long term—for example, how economic preferences affect human capital formation, household production, consumption, within-household behavior, and between-peer behavior—would be valuable. Such work would need to be longitudinal in nature and would be valuable to understanding decisions over the life course. This would lead policy makers and researchers to place more value on economic preferences. Another open question is how preferences

affect well-being. This does not mean that intermediate-term outcomes should be ignored; indeed, research suggests that collecting multiple intermediate outcomes can be helpful for drawing inference about long-term outcomes (e.g., Athey et al. 2019).

Second, few papers gather causal evidence. Instead, much of the work correlates SES or parenting with economic preferences. This correlational work is important, but causal inference demands either natural or field experiments to assess how investments in children affect preferences. It is also instructive to explore heterogeneous treatment effects, since how children respond to an intervention might itself be affected by preferences. Interventions designed to affect a skill or welfare (such as early childhood programs) may provide benefits in that area, but may also have unintended consequences on preferences. Understanding the consequences on preferences (as is done in Cappelen et al. 2020) is valuable, since if policies shape preferences, then they ultimately shape institutions as children become adults. Another open question is whether there are “critical periods” to intervene with children. Conducting experiments that exogenously manipulate the environment at different ages would allow researchers to answer such questions. Finally, we pointed out in this paper that children can be considered “inexperienced” subjects in many ways, hence they can be used for further study to understand how experience affects decision making, as in List (2004a).

The field is relatively new, and therefore a common methodology for conducting experiments with children has not yet been established. While table 1 identifies what methods most papers use in each age range, there are a number of papers that diverge from the most common—and therefore potentially from the most appropriate—method. It is our hope that this paper advances our understanding of how best to conduct experiments

with children and adolescents, and provides a common set of guidelines that allow experimental designs to become more consistent with one another. This will allow for more direct comparability in findings.

The field has had a historic lack of discussion with the sister sciences. Tools from psychology—such as using multiple measures and assessing test–retest reliability—can be helpful in advancing our understanding of how preference measurements should be collected. Multiple measures are especially important in cases where we expect error, and economists have thought about this issue (e.g., Heckman, Pinto, and Savelyev 2013; Gillen, Snowberg, and Yarov 2019). Collecting data on how other fields measure similar constructs (such as using stated-preference questionnaires for risk or time) can be valuable for designing multiple measures. Alternatively, economists also have value to add to the sister sciences. For instance, economists use incentivized, context-free elicitation tasks that differ significantly from those used in the sister sciences. As such, a number of papers in the psychology literature already use economic games such as dictator games. Economists are also well trained in quantitative methods, and can impart knowledge about power, multiple hypothesis testing and related concepts to researchers in other fields.

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